

Family-Based Association: PLINK, FaST-LMM, and GCTA

In this exercise you will carry out association analysis of data from a mini genome-wide association study. The data comes from families (related individuals) measured for a quantitative trait of interest. The purpose is to detect which (if any) of the loci are associated with the quantitative trait.

Methodology: we compare standard linear regression (PLINK) against two linear mixed model approaches that account for relatedness between individuals: **FaST-LMM** and **GCTA**. For GCTA we also estimate the heritability accounted for by all genotyped SNPs, and by various subsets of SNPs.

Data overview

Family data consisting of 498 individuals typed at 134,946 SNPs. All individuals have measurements of a quantitative trait of interest; the data has already been QC-ed.

The data is in PLINK binary-file format under `data/`: `quantfamdata.bed` (genotypes), `quantfamdata.bim` (map file), and `quantfamdata.fam` (pedigree structure — the last column gives each individual's quantitative trait value). `data/myphenos.txt` is a ready-made phenotype file derived from `quantfamdata.fam` columns 1, 2, and 6 (family ID, individual ID, trait value) — this is what `--pheno` points to throughout.

How this notebook is organized

Part 1 (Bash kernel) runs every association and heritability analysis end-to-end, producing all the result files used in Part 2:

1. **PLINK** — standard linear regression, ignoring relatedness.
2. **FaST-LMM** — a linear mixed model that accounts for relatedness.
3. **GCTA** `--mlma` — GCTA's own mixed-model association test.
4. **GCTA GREML** — SNP heritability estimation: genome-wide, naively per-chromosome, and then properly partitioned across all 22 autosomes.
5. **GCTA fastGWA** — GCTA's sparse-GRM mixed model, designed for biobank-scale data.

Part 2 (R kernel) reads the result files from sections 1, 2, 3, and 5 above and produces QQ plots, Manhattan plots, and genomic-control (λ) checks for each, plus a direct comparison between GCTA `--mlma` and FaST-LMM, and between GCTA fastGWA and

GCTA `--mlma` . (The GREML heritability estimates from section 4 are read directly off the GCTA log output within Part 1 — there's nothing further to plot for those in Part 2.) You'll switch kernels **once**, right before Part 2 — in JupyterLab's menu bar: **Kernel** → **Change Kernel...** → **R**, then continue running cells from there.

Program documentation

- **PLINK**: [1.90 docs](#)
- **R**: [r-project.org](#) — from within R, `help()` documents any command
- **FaST-LMM**: [fastlmm.github.io](#) while a newer Python `fastlmm` package is available, this notebook instead runs the original `fastlmmc` binary for the exercise.
- **GCTA**: [cnsgenomics.com/software/gcta](#)

References

- Chang CC, Chow CC, Tellier LCAM, Vattikuti S, Purcell SM, Lee JJ (2015) Second-generation PLINK: rising to the challenge of larger and richer datasets. *GigaScience*, 4.
 - Lippert C, Listgarten J, Liu Y, Kadie CM, Davidson RI, Heckerman D (2011) FaST linear mixed models for genome-wide association studies. *Nat Methods* 8(10):833–835.
 - Purcell S, Neale B, Todd-Brown K, Thomas L, Ferreira MAR, Bender D, Maller J, Sklar P, de Bakker PIW, Daly MJ & Sham PC (2007) PLINK: a toolset for whole-genome association and population-based linkage analysis. *American Journal of Human Genetics*, 81:559–575.
 - R Core Team (2025), R: A Language and Environment for Statistical Computing, R Foundation for Statistical Computing
 - Yang et al. (2011) GCTA: A tool for genome-wide complex trait analysis. *American Journal of Human Genetics*, 88:76–82.
-

Part 1 — Running the Analyses (Bash kernel)

Everything in this part uses the **Bash** kernel. Each step writes its results to a file, which Part 2 reads back in with R — nothing here depends on R having run first.

Kernel: run the code cells in this section with the **Bash** kernel.

1. PLINK: standard linear regression

To start with, we use PLINK to perform a test equivalent to linear regression analysis, without worrying about the relatedness between individuals. Standard regression assumes each observation is independent — an assumption that's violated here, since this is family data with related individuals who share genetic background. That violation can inflate test statistics and produce false-positive associations, which is exactly what the mixed-model approaches in the rest of this notebook are designed to correct for; PLINK's results here are the "naive" baseline everything else gets compared against.

A copy of the screen output is saved to `plinkresults.log`; the association results themselves go to `plinkresults.qassoc`, one line per SNP, with columns `CHR`, `SNP`, `BP`, `NMISS`, `BETA`, `SE`, `R2`, `T`, and `P` (the most useful being `T`, the test statistic, and `P`, its p-value).

```
In [1]: mkdir -p output

ls -l ./data # sanity check: confirm the input PLINK files are where we expect
# --assoc: simple allelic association test (no relatedness or covariates model)
plink --bfile ./data/quantfamdata --assoc --out output/plinkresults
```

```

total 20180
-rw----- 1 root root      432 Jul 21 20:07 multipleGRMs.txt
-rw----- 1 root root       38 Jul 21 20:07 multipleGRMs6andnot6.txt
-rw----- 1 root root      413 Jul 21 20:07 multipleGRMsnot6.txt
-rw----- 1 root root     9627 Jul 21 19:14 myphenos.txt
-rw----- 1 root root 16868253 Jul 21 19:14 quantfamdata.bed
-rw----- 1 root root 3751046 Jul 21 19:14 quantfamdata.bim
-rw----- 1 root root   14938 Jul 21 19:14 quantfamdata.fam
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)                                cog-genomics.org/plink/1.
9/
(C) 2005-2024 Shaun Purcell, Christopher Chang    GNU General Public License
v3
Logging to output/plinkresults.log.
Options in effect:
  --assoc
  --bfile ./data/quantfamdata
  --out output/plinkresults

7836 MB RAM detected; reserving 3918 MB for main workspace.
134946 variants loaded from .bim file.
498 people (226 males, 272 females) loaded from .fam.
498 phenotype values loaded from .fam.
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 124 founders and 374 nonfounders present.
Calculating allele frequencies... 101112131415161718192021222324252627282930
3132333435363738394041424344454647484950515253545556575859606162636465666768
69707172737475767778798081828384858687888990919293949596979899 done.
Total genotyping rate is 0.999457.
134946 variants and 498 people pass filters and QC.
Phenotype data is quantitative.
Writing QT --assoc report to output/plinkresults.qassoc ... 1011121314151617
1819202122232425262728293031323334353637383940414243444546474849505152535455
5657585960616263646566676869707172737475767778798081828384858687888990919293
949596979899done.

```

2. FaST-LMM: linear mixed model accounting for relatedness

Now we re-run the analysis using FaST-LMM, which estimates and accounts for relatedness between individuals.

We use the `--bfile quantfamdata` command to tell the program the name (stem) of the files with the input genotype data containing the SNPs to be tested for association, and the `--bfileSim quantfamdata` command to tell the program the name of the files containing the SNPs to be used for estimating relatedness. Here we just use the same files both times, but FaST-LMM would allow us to use different files for these two operations if we prefer.

The command `--pheno` reads the phenotype column from `data/myphenos.txt`. Results are written to `output/FLMMresults`, already ordered by significance, with columns `SNP`, `Chromosome`, `Position`, `Pvalue` (among others).

```
In [2]: # Runs the actual fastlmmc binary; -bfileSim gives the genotype data used to
fastlmmc -bfile ./data/quantfamdata -pheno ./data/myphenos.txt -bfileSim ./c
```

FastLmmC v2.07.20260731 - Factored Spectrally Transformed Linear Mixed Models [Release]

Copyright Microsoft Corporation -- Licensed Only for Non-Commercial use.
Compiled Jul 31 2026 at 17:20:32 by 6beff924acff for Linux v3.x kernel
using OpenBLAS

```
++ Start Processing CommandLine:
-- End Processing CommandLine:

++ Start Loading FastLmm Data:
++ Start Loading Covariance Data:
++ Processing PLINK fileset: [./data/quantfamdata]
ReadBinaryFiles4()* elapsed time: 441.944 ms
  Number of Individuals Selected: 498
    Number of Phenotypes: 1
    Number of SNPs Read: 134946
    Number of SNPs Used: 134946
-- End Processing PLINK fileset: [./data/quantfamdata]
-- End Loading Covariance Data:
++ Start Loading Test Data:
++ Processing PLINK fileset: [./data/quantfamdata]
ReadBinaryFiles4()* elapsed time: 476.870 ms
  Number of Individuals Selected: 498
    Number of Phenotypes: 1
    Number of SNPs Read: 134946
    Number of SNPs Used: 134946
-- End Processing PLINK fileset: [./data/quantfamdata]
-- End Loading Test Data:

-- End Loading FastLmm Data:

    Compute/Load EigenSym:
    Compute GWAS using LMM:
      GWAS elapsed time: 16.871 sec
      Write output file: [/home/jupyter/output/FLMMresults]
      Total elapsed time: 20.352 sec
```

3. GCTA `--mlma`: GCTA's own mixed-model association test

GCTA is designed primarily for apparently-unrelated individuals, but we apply it here to our related individuals for comparison with FaST-LMM. `--mlma` performs mixed-linear-model association analysis, fitting the same kind of genetic random effect described above; `--bfile` and `--pheno` give the genotype and phenotype data (`data/myphenos.txt` , derived from `quantfamdata.fam` as described above).

```
In [3]: # --mlma: mixed-linear-model association test (GCTA estimates relatedness in
gcta64 --mlma --bfile ./data/quantfamdata --pheno ./data/myphenos.txt --out
```

```

*****
* Genome-wide Complex Trait Analysis (GCTA)
* version v1.95.3 Linux
* Built at Sep  9 2026 10:56:04, by GCC 12.2
* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
*****
Analysis started at 05:40:45 UTC on Thu Sep 10 2026.
Hostname: 8768ca282dde

```

Accepted options:

```

--mlma
--bfile ./data/quantfamdata
--pheno ./data/myphenos.txt
--out output/GCTAresults

```

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

```

Reading PLINK FAM file from [./data/quantfamdata.fam].
498 individuals to be included from [./data/quantfamdata.fam].
Reading PLINK BIM file from [./data/quantfamdata.bim].
134946 SNPs to be included from [./data/quantfamdata.bim].
Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format
...
Genotype data for 498 individuals and 134946 SNPs to be included from [./data/quantfamdata.bed].
Reading phenotypes from [./data/myphenos.txt].
Non-missing phenotypes of 498 individuals are included from [./data/myphenos.txt].
Calculating allele frequencies ...
Recoding genotypes (individual major mode) ...

Calculating the genetic relationship matrix (GRM) ...
498 individuals are in common in these files.

```

Performing MLM association analyses (including the candidate SNP) ...

```

Performing REML analysis ... (Note: may take hours depending on sample size).
498 observations, 1 fixed effect(s), and 2 variance component(s)(including residual variance).
Calculating prior values of variance components by EM-REML ...
Updated prior values: 31.0444 31.1001
logL: -1272.01
Running AI-REML algorithm ...

```

Iter.	logL	V(G)	V(e)
1	-1272.00	30.59135	31.46507
2	-1272.00	29.65726	32.21513
3	-1271.99	29.74705	32.13462
4	-1271.99	29.73839	32.14262
5	-1271.99	29.73924	32.14184

```

Log-likelihood ratio converged.

```

Running association tests for 134946 SNPs ...

Saving the results of the mixed linear model association analyses of 134946 SNPs to [output/GCTAresults.mlma] ...

Analysis finished at 05:42:05 UTC on Thu Sep 10 2026
Overall computational time: 1 minute 20 sec.

4. GCTA: SNP heritability (GREML)

GCTA can be used to estimate the heritability accounted for by all autosomal genome-wide SNPs.

To estimate the heritability, we first build a genetic relationship matrix (GRM) — from all the autosomal significant SNPs. We then the GREML method (genomic-relatedness-based REML) to fit a mixed model with the GRM as a random effect and estimate the (SNP) heritability.

```
In [4]: # 1. Build the genetic relationship matrix (GRM) from all autosomal SNPs
gcta64 --bfile ./data/quantfamdata --autosome --make-grm-bin --out output/GC
# 2. Estimate heritability (REML) using that GRM
gcta64 --reml --grm-bin output/GCTAgrm --pheno ./data/myphenos.txt --out out
```

```
*****
* Genome-wide Complex Trait Analysis (GCTA)
* version v1.95.3 Linux
* Built at Sep  9 2026 10:56:04, by GCC 12.2
* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
*****
Analysis started at 05:42:05 UTC on Thu Sep 10 2026.
Hostname: 8768ca282dde
```

Accepted options:
--bfile ./data/quantfamdata
--autosome
--make-grm-bin
--out output/GCTAgrm

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

Reading PLINK FAM file from [./data/quantfamdata.fam].
498 individuals to be included from [./data/quantfamdata.fam].
Reading PLINK BIM file from [./data/quantfamdata.bim].
134946 SNPs to be included from [./data/quantfamdata.bim].
134946 SNPs from chromosome 1 to chromosome 22 are included in the analysis.
Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format
...
Genotype data for 498 individuals and 134946 SNPs to be included from [./data/quantfamdata.bed].
Calculating allele frequencies ...
Recoding genotypes (individual major mode) ...

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)

Summary of the GRM:
Mean of diagonals = 1.00289
Variance of diagonals = 0.000422079
Mean of off-diagonals = -0.00201898
Variance of off-diagonals = 0.0004337
GRM of 498 individuals has been saved in the file [output/GCTAgrm.grm.bin] (in binary format).
Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrm.grm.N.bin] (in binary format).
IDs for the GRM file [output/GCTAgrm.grm.bin] have been saved in the file [output/GCTAgrm.grm.id].

Analysis finished at 05:42:08 UTC on Thu Sep 10 2026
Overall computational time: 2.66 sec.

```
*****
* Genome-wide Complex Trait Analysis (GCTA)
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* Built at Sep  9 2026 10:56:04, by GCC 12.2
* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
```

Analysis started at 05:42:08 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

--reml
--grm-bin output/GCTAgrm
--pheno ./data/myphenos.txt
--out output/GCTAherit

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

Reading IDs of the GRM from [output/GCTAgrm.grm.id].

498 IDs are read from [output/GCTAgrm.grm.id].

Reading the GRM from [output/GCTAgrm.grm.bin].

GRM for 498 individuals are included from [output/GCTAgrm.grm.bin].

Reading phenotypes from [./data/myphenos.txt].

Non-missing phenotypes of 498 individuals are included from [./data/myphenos.txt].

498 individuals are in common in these files.

Performing REML analysis ... (Note: may take hours depending on sample size).

498 observations, 1 fixed effect(s), and 2 variance component(s)(including residual variance).

Calculating prior values of variance components by EM-REML ...

Updated prior values: 31.0442 31.1

logL: -1272.01

Running AI-REML algorithm ...

Iter.	logL	V(G)	V(e)
1	-1272.00	30.59118	31.46479
2	-1272.00	29.65709	32.21465
3	-1271.99	29.74688	32.13414
4	-1271.99	29.73822	32.14214
5	-1271.99	29.73907	32.14136

Log-likelihood ratio converged.

Calculating the logLikelihood for the reduced model ...

(variance component 1 is dropped from the model)

Calculating prior values of variance components by EM-REML ...

Updated prior values: 62.30497

logL: -1278.41752

Running AI-REML algorithm ...

Iter.	logL	V(e)
1	-1278.42	62.30497

Log-likelihood ratio converged.

Summary result of REML analysis:

Source	Variance	SE
V(G)	29.739073	9.571995
V(e)	32.141356	8.886473
Vp	61.880429	3.993645
V(G)/Vp	0.480589	0.145944

Sampling variance/covariance of the estimates of variance components:

```
9.162309e+01    -7.732164e+01
-7.732164e+01    7.896940e+01
```

Summary result of REML analysis has been saved in the file [output/GCTAherit.hsq].

Analysis finished at 05:42:08 UTC on Thu Sep 10 2026

Overall computational time: 0.06 sec.

Per-chromosome heritability (naive)

Repeating this separately for chromosomes 1, 2, 6, and 12: expect chromosomes 1 and 2 to look non-significant given their standard errors, while 6 and 12 (where the earlier association signals were) look significant. Note the four chromosome-specific estimates sum to *more* than the overall heritability — with related individuals, single-chromosome GRMs can be confounded by shared non-genetic effects or correlations with other chromosomes, inflating each chromosome's apparent contribution.

```
In [5]: # Chromosome 1
gcta64 --bfile ./data/quantfamdata --chr 1 --make-grm-bin --out output/GCTA
gcta64 --reml --grm-bin output/GCTAgrmchr1 --pheno ./data/myphenos.txt \
--out output/GCTAheritchr1

# Chromosome 2
gcta64 --bfile ./data/quantfamdata --chr 2 --make-grm-bin --out output/GCTA
gcta64 --reml --grm-bin output/GCTAgrmchr2 --pheno ./data/myphenos.txt \
--out output/GCTAheritchr2

# Chromosome 6
gcta64 --bfile ./data/quantfamdata --chr 6 --make-grm-bin --out output/GCTA
gcta64 --reml --grm-bin output/GCTAgrmchr6 --pheno ./data/myphenos.txt --out

# Chromosome 12
gcta64 --bfile ./data/quantfamdata --chr 12 --make-grm-bin --out output/GCTA
gcta64 --reml --grm-bin output/GCTAgrmchr12 --pheno ./data/myphenos.txt --ou
```

```
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* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
*****
Analysis started at 05:42:08 UTC on Thu Sep 10 2026.
Hostname: 8768ca282dde
```

```
Accepted options:
--bfile ./data/quantfamdata
--chr 1
--make-grm-bin
--out output/GCTAgrmchr1
```

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

```
Reading PLINK FAM file from [./data/quantfamdata.fam].
498 individuals to be included from [./data/quantfamdata.fam].
Reading PLINK BIM file from [./data/quantfamdata.bim].
134946 SNPs to be included from [./data/quantfamdata.bim].
9003 SNPs on chromosome 1 are included in the analysis.
Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format
...
Genotype data for 498 individuals and 9003 SNPs to be included from [./data/quantfamdata.bed].
Calculating allele frequencies ...
Recoding genotypes (individual major mode) ...
```

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)

```
Summary of the GRM:
Mean of diagonals = 1.00293
Variance of diagonals = 0.0015773
Mean of off-diagonals = -0.0020191
Variance of off-diagonals = 0.000981762
GRM of 498 individuals has been saved in the file [output/GCTAgrmchr1.grm.bin] (in binary format).
Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrmchr1.grm.N.bin] (in binary format).
IDs for the GRM file [output/GCTAgrmchr1.grm.bin] have been saved in the file [output/GCTAgrmchr1.grm.id].
```

```
Analysis finished at 05:42:09 UTC on Thu Sep 10 2026
Overall computational time: 0.38 sec.
```

```
*****
* Genome-wide Complex Trait Analysis (GCTA)
* version v1.95.3 Linux
* Built at Sep  9 2026 10:56:04, by GCC 12.2
* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
```

Analysis started at 05:42:09 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

--reml
--grm-bin output/GCTAgrmchr1
--pheno ./data/myphenos.txt
--out output/GCTAheritchr1

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

Reading IDs of the GRM from [output/GCTAgrmchr1.grm.id].

498 IDs are read from [output/GCTAgrmchr1.grm.id].

Reading the GRM from [output/GCTAgrmchr1.grm.bin].

GRM for 498 individuals are included from [output/GCTAgrmchr1.grm.bin].

Reading phenotypes from [./data/myphenos.txt].

Non-missing phenotypes of 498 individuals are included from [./data/myphenos.txt].

498 individuals are in common in these files.

Performing REML analysis ... (Note: may take hours depending on sample size).

498 observations, 1 fixed effect(s), and 2 variance component(s)(including residual variance).

Calculating prior values of variance components by EM-REML ...

Updated prior values: 31.1649 32.7147

logL: -1280.1

Running AI-REML algorithm ...

Iter.	logL	V(G)	V(e)
1	-1279.59	25.51133	37.48693
2	-1278.14	13.16935	48.55726
3	-1276.74	13.08769	48.83873
4	-1276.74	13.02830	49.03582
5	-1276.73	12.89243	49.47118
6	-1276.73	12.87786	49.48644
7	-1276.73	12.87688	49.48735

Log-likelihood ratio converged.

Calculating the logLikelihood for the reduced model ...

(variance component 1 is dropped from the model)

Calculating prior values of variance components by EM-REML ...

Updated prior values: 62.30497

logL: -1278.41752

Running AI-REML algorithm ...

Iter.	logL	V(e)
1	-1278.42	62.30497

Log-likelihood ratio converged.

Summary result of REML analysis:

Source	Variance	SE
V(G)	12.876883	7.277044
V(e)	49.487348	7.395969

Vp 62.364231 3.992076
V(G)/Vp 0.206479 0.114179

Sampling variance/covariance of the estimates of variance components:
5.295536e+01 -4.585953e+01
-4.585953e+01 5.470036e+01

Summary result of REML analysis has been saved in the file [output/GCTAherit
chr1.hsq].

Analysis finished at 05:42:09 UTC on Thu Sep 10 2026

Overall computational time: 0.07 sec.

* Genome-wide Complex Trait Analysis (GCTA)

* version v1.95.3 Linux

* Built at Sep 9 2026 10:56:04, by GCC 12.2

* (C) 2010-present, Yang Lab, Westlake University

* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>

Analysis started at 05:42:09 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

--bfile ./data/quantfamdata

--chr 2

--make-grm-bin

--out output/GCTAgrmchr2

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

Reading PLINK FAM file from [./data/quantfamdata.fam].

498 individuals to be included from [./data/quantfamdata.fam].

Reading PLINK BIM file from [./data/quantfamdata.bim].

134946 SNPs to be included from [./data/quantfamdata.bim].

9721 SNPs on chromosome 2 are included in the analysis.

Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format

...

Genotype data for 498 individuals and 9721 SNPs to be included from [./data/quantfamdata.bed].

Calculating allele frequencies ...

Recoding genotypes (individual major mode) ...

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)

Summary of the GRM:

Mean of diagonals = 1.00164

Variance of diagonals = 0.00203532

Mean of off-diagonals = -0.00201645

Variance of off-diagonals = 0.00104589

GRM of 498 individuals has been saved in the file [output/GCTAgrmchr2.grm.bin] (in binary format).

Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrmchr2.grm.N.bin] (in binary format).

ry format).

IDs for the GRM file [output/GCTAgrmchr2.grm.bin] have been saved in the file [output/GCTAgrmchr2.grm.id].

Analysis finished at 05:42:09 UTC on Thu Sep 10 2026

Overall computational time: 0.40 sec.

* Genome-wide Complex Trait Analysis (GCTA)

* version v1.95.3 Linux

* Built at Sep 9 2026 10:56:04, by GCC 12.2

* (C) 2010-present, Yang Lab, Westlake University

* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>

Analysis started at 05:42:09 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

--reml

--grm-bin output/GCTAgrmchr2

--pheno ./data/myphenos.txt

--out output/GCTAheritchr2

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

Reading IDs of the GRM from [output/GCTAgrmchr2.grm.id].

498 IDs are read from [output/GCTAgrmchr2.grm.id].

Reading the GRM from [output/GCTAgrmchr2.grm.bin].

GRM for 498 individuals are included from [output/GCTAgrmchr2.grm.bin].

Reading phenotypes from [./data/myphenos.txt].

Non-missing phenotypes of 498 individuals are included from [./data/myphenos.txt].

498 individuals are in common in these files.

Performing REML analysis ... (Note: may take hours depending on sample size).

498 observations, 1 fixed effect(s), and 2 variance component(s)(including residual variance).

Calculating prior values of variance components by EM-REML ...

Updated prior values: 31.1 33.381

logL: -1284.45

Running AI-REML algorithm ...

Iter.	logL	V(G)	V(e)
1	-1283.43	22.49448	40.29567
2	-1280.34	16.97034	45.18315
3	-1278.89	13.46291	48.51428
4	-1278.24	11.22354	50.75247
5	-1277.95	6.64600	55.49657
6	-1277.71	6.94925	55.33098
7	-1277.71	6.94493	55.33588
8	-1277.71	6.94504	55.33577

Log-likelihood ratio converged.

Calculating the logLikelihood for the reduced model ...

```
(variance component 1 is dropped from the model)
Calculating prior values of variance components by EM-REML ...
Updated prior values: 62.30497
logL: -1278.41752
Running AI-REML algorithm ...
Iter.   logL      V(e)
1       -1278.42    62.30497
Log-likelihood ratio converged.
```

```
Summary result of REML analysis:
Source  Variance      SE
V(G)    6.945039      6.391182
V(e)    55.335774      7.029114
Vp      62.280813      3.961202
V(G)/Vp 0.111512      0.101862
```

```
Sampling variance/covariance of the estimates of variance components:
4.084720e+01   -3.728226e+01
-3.728226e+01   4.940844e+01
```

Summary result of REML analysis has been saved in the file [output/GCTAherit chr2.hsq].

Analysis finished at 05:42:09 UTC on Thu Sep 10 2026
Overall computational time: 0.07 sec.

```
*****
* Genome-wide Complex Trait Analysis (GCTA)
* version v1.95.3 Linux
* Built at Sep  9 2026 10:56:04, by GCC 12.2
* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
*****
```

Analysis started at 05:42:10 UTC on Thu Sep 10 2026.
Hostname: 8768ca282dde

```
Accepted options:
--bfile ./data/quantfamdata
--chr 6
--make-grm-bin
--out output/GCTAgrmchr6
```

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

```
Reading PLINK FAM file from [./data/quantfamdata.fam].
498 individuals to be included from [./data/quantfamdata.fam].
Reading PLINK BIM file from [./data/quantfamdata.bim].
134946 SNPs to be included from [./data/quantfamdata.bim].
19811 SNPs on chromosome 6 are included in the analysis.
Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format
...
```

```
Genotype data for 498 individuals and 19811 SNPs to be included from [./data/quantfamdata.bed].
Calculating allele frequencies ...
Recoding genotypes (individual major mode) ...
```

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)

Summary of the GRM:

Mean of diagonals = 1.00362

Variance of diagonals = 0.00196349

Mean of off-diagonals = -0.00202046

Variance of off-diagonals = 0.00121764

GRM of 498 individuals has been saved in the file [output/GCTAgrmchr6.grm.bin] (in binary format).

Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrmchr6.grm.N.bin] (in binary format).

IDs for the GRM file [output/GCTAgrmchr6.grm.bin] have been saved in the file [output/GCTAgrmchr6.grm.id].

Analysis finished at 05:42:10 UTC on Thu Sep 10 2026

Overall computational time: 0.63 sec.

* Genome-wide Complex Trait Analysis (GCTA)

* version v1.95.3 Linux

* Built at Sep 9 2026 10:56:04, by GCC 12.2

* (C) 2010-present, Yang Lab, Westlake University

* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>

Analysis started at 05:42:10 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

--reml

--grm-bin output/GCTAgrmchr6

--pheno ./data/myphenos.txt

--out output/GCTAheritchr6

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

Reading IDs of the GRM from [output/GCTAgrmchr6.grm.id].

498 IDs are read from [output/GCTAgrmchr6.grm.id].

Reading the GRM from [output/GCTAgrmchr6.grm.bin].

GRM for 498 individuals are included from [output/GCTAgrmchr6.grm.bin].

Reading phenotypes from [./data/myphenos.txt].

Non-missing phenotypes of 498 individuals are included from [./data/myphenos.txt].

498 individuals are in common in these files.

Performing REML analysis ... (Note: may take hours depending on sample size).

498 observations, 1 fixed effect(s), and 2 variance component(s)(including residual variance).

Calculating prior values of variance components by EM-REML ...

Updated prior values: 31.0069 31.947

logL: -1271.19


```
Running AI-REML algorithm ...
Iter.   logL      V(G)      V(e)
1       -1271.04      28.12821      34.35332
2       -1270.59      22.24402      39.49029
3       -1270.25      22.86423      39.12826
4       -1270.25      22.81962      39.16883
5       -1270.25      22.82373      39.16530
6       -1270.25      22.82336      39.16562
Log-likelihood ratio converged.
```

```
Calculating the logLikelihood for the reduced model ...
(variance component 1 is dropped from the model)
Calculating prior values of variance components by EM-REML ...
Updated prior values: 62.30497
logL: -1278.41752
Running AI-REML algorithm ...
Iter.   logL      V(e)
1       -1278.42      62.30497
Log-likelihood ratio converged.
```

```
Summary result of REML analysis:
Source  Variance      SE
V(G)    22.823363      6.974969
V(e)    39.165615      6.391106
Vp      61.988978      4.041888
V(G)/Vp 0.368184      0.104517
```

```
Sampling variance/covariance of the estimates of variance components:
4.865019e+01   -3.657978e+01
-3.657978e+01   4.084623e+01
```

Summary result of REML analysis has been saved in the file [output/GCTAherit chr6.hsqr].

Analysis finished at 05:42:10 UTC on Thu Sep 10 2026
Overall computational time: 0.10 sec.

```
*****
* Genome-wide Complex Trait Analysis (GCTA)
* version v1.95.3 Linux
* Built at Sep  9 2026 10:56:04, by GCC 12.2
* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
*****
```

Analysis started at 05:42:11 UTC on Thu Sep 10 2026.
Hostname: 8768ca282dde

```
Accepted options:
--bfile ./data/quantfamdata
--chr 12
--make-grm-bin
--out output/GCTAgrmchr12
```

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

Reading PLINK FAM file from [./data/quantfamdata.fam].
498 individuals to be included from [./data/quantfamdata.fam].
Reading PLINK BIM file from [./data/quantfamdata.bim].
134946 SNPs to be included from [./data/quantfamdata.bim].
18385 SNPs on chromosome 12 are included in the analysis.
Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format
...
Genotype data for 498 individuals and 18385 SNPs to be included from [./data/quantfamdata.bed].
Calculating allele frequencies ...
Recoding genotypes (individual major mode) ...

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)

Summary of the GRM:
Mean of diagonals = 1.00351
Variance of diagonals = 0.00188967
Mean of off-diagonals = -0.00202017
Variance of off-diagonals = 0.00122423
GRM of 498 individuals has been saved in the file [output/GCTAgrmchr12.grm.bin] (in binary format).
Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrmchr12.grm.N.bin] (in binary format).
IDs for the GRM file [output/GCTAgrmchr12.grm.bin] have been saved in the file [output/GCTAgrmchr12.grm.id].

Analysis finished at 05:42:11 UTC on Thu Sep 10 2026

Overall computational time: 0.63 sec.

* Genome-wide Complex Trait Analysis (GCTA)
* version v1.95.3 Linux
* Built at Sep 9 2026 10:56:04, by GCC 12.2
* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>

Analysis started at 05:42:11 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

--reml
--grm-bin output/GCTAgrmchr12
--pheno ./data/myphenos.txt
--out output/GCTAheritchr12

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

Reading IDs of the GRM from [output/GCTAgrmchr12.grm.id].
498 IDs are read from [output/GCTAgrmchr12.grm.id].
Reading the GRM from [output/GCTAgrmchr12.grm.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr12.grm.bin].
Reading phenotypes from [./data/myphenos.txt].
Non-missing phenotypes of 498 individuals are included from [./data/myphenos.txt].

s.txt].

498 individuals are in common in these files.

Performing REML analysis ... (Note: may take hours depending on sample size).

498 observations, 1 fixed effect(s), and 2 variance component(s)(including residual variance).

Calculating prior values of variance components by EM-REML ...

Updated prior values: 31.0486 32.5394

logL: -1276.3

Running AI-REML algorithm ...

Iter.	logL	V(G)	V(e)
1	-1275.89	26.47687	36.32801
2	-1274.80	17.05759	44.64973
3	-1273.93	17.31114	44.56457
4	-1273.92	17.85561	44.38722
5	-1273.91	17.83594	44.40740
6	-1273.91	17.83714	44.40633

Log-likelihood ratio converged.

Calculating the logLikelihood for the reduced model ...

(variance component 1 is dropped from the model)

Calculating prior values of variance components by EM-REML ...

Updated prior values: 62.30497

logL: -1278.41752

Running AI-REML algorithm ...

Iter.	logL	V(e)
1	-1278.42	62.30497

Log-likelihood ratio converged.

Summary result of REML analysis:

Source	Variance	SE
V(G)	17.837137	6.876731
V(e)	44.406334	6.655022
Vp	62.243471	4.019753
V(G)/Vp	0.286570	0.105506

Sampling variance/covariance of the estimates of variance components:

4.728943e+01	-3.771017e+01
-3.771017e+01	4.428932e+01

Summary result of REML analysis has been saved in the file [output/GCTAherit chr12.hsq].

Analysis finished at 05:42:11 UTC on Thu Sep 10 2026

Overall computational time: 0.14 sec.

Correctly partitioning heritability across all 22 autosomes

To partition heritability properly we need chromosome-specific GRMs for every autosome, included simultaneously in one model — not estimated separately as above.

First build the remaining per-chromosome GRMs:

```
In [6]: # Same two-step (build-GRM) pattern as above, repeated for each remaining au
# (chromosomes 1, 2, 6, and 12 were already built above)
gcta64 --bfile ./data/quantfamdata --chr 3 --make-grm-bin --out output/GCTA
gcta64 --bfile ./data/quantfamdata --chr 4 --make-grm-bin --out output/GCTA
gcta64 --bfile ./data/quantfamdata --chr 5 --make-grm-bin --out output/GCTA
gcta64 --bfile ./data/quantfamdata --chr 7 --make-grm-bin --out output/GCTA
gcta64 --bfile ./data/quantfamdata --chr 8 --make-grm-bin --out output/GCTA
gcta64 --bfile ./data/quantfamdata --chr 9 --make-grm-bin --out output/GCTA
gcta64 --bfile ./data/quantfamdata --chr 10 --make-grm-bin --out output/GCTA
gcta64 --bfile ./data/quantfamdata --chr 11 --make-grm-bin --out output/GCTA
gcta64 --bfile ./data/quantfamdata --chr 13 --make-grm-bin --out output/GCTA
gcta64 --bfile ./data/quantfamdata --chr 14 --make-grm-bin --out output/GCTA
gcta64 --bfile ./data/quantfamdata --chr 15 --make-grm-bin --out output/GCTA
gcta64 --bfile ./data/quantfamdata --chr 16 --make-grm-bin --out output/GCTA
gcta64 --bfile ./data/quantfamdata --chr 17 --make-grm-bin --out output/GCTA
gcta64 --bfile ./data/quantfamdata --chr 18 --make-grm-bin --out output/GCTA
gcta64 --bfile ./data/quantfamdata --chr 19 --make-grm-bin --out output/GCTA
gcta64 --bfile ./data/quantfamdata --chr 20 --make-grm-bin --out output/GCTA
gcta64 --bfile ./data/quantfamdata --chr 21 --make-grm-bin --out output/GCTA
gcta64 --bfile ./data/quantfamdata --chr 22 --make-grm-bin --out output/GCTA
```

```
*****
* Genome-wide Complex Trait Analysis (GCTA)
* version v1.95.3 Linux
* Built at Sep  9 2026 10:56:04, by GCC 12.2
* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
*****
Analysis started at 05:42:12 UTC on Thu Sep 10 2026.
Hostname: 8768ca282dde
```

```
Accepted options:
--bfile ./data/quantfamdata
--chr 3
--make-grm-bin
--out output/GCTAgrmchr3
```

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

```
Reading PLINK FAM file from [./data/quantfamdata.fam].
498 individuals to be included from [./data/quantfamdata.fam].
Reading PLINK BIM file from [./data/quantfamdata.bim].
134946 SNPs to be included from [./data/quantfamdata.bim].
8040 SNPs on chromosome 3 are included in the analysis.
Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format
...
Genotype data for 498 individuals and 8040 SNPs to be included from [./data/quantfamdata.bed].
Calculating allele frequencies ...
Recoding genotypes (individual major mode) ...
```

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)

```
Summary of the GRM:
Mean of diagonals = 1.00364
Variance of diagonals = 0.00208634
Mean of off-diagonals = -0.00202046
Variance of off-diagonals = 0.00115896
GRM of 498 individuals has been saved in the file [output/GCTAgrmchr3.grm.bin] (in binary format).
Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrmchr3.grm.N.bin] (in binary format).
IDs for the GRM file [output/GCTAgrmchr3.grm.bin] have been saved in the file [output/GCTAgrmchr3.grm.id].
```

```
Analysis finished at 05:42:12 UTC on Thu Sep 10 2026
Overall computational time: 0.42 sec.
```

```
*****
* Genome-wide Complex Trait Analysis (GCTA)
* version v1.95.3 Linux
* Built at Sep  9 2026 10:56:04, by GCC 12.2
* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
```

Analysis started at 05:42:12 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

--bfile ./data/quantfamdata

--chr 4

--make-grm-bin

--out output/GCTAgrmchr4

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

Reading PLINK FAM file from [./data/quantfamdata.fam].

498 individuals to be included from [./data/quantfamdata.fam].

Reading PLINK BIM file from [./data/quantfamdata.bim].

134946 SNPs to be included from [./data/quantfamdata.bim].

7734 SNPs on chromosome 4 are included in the analysis.

Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format

...

Genotype data for 498 individuals and 7734 SNPs to be included from [./data/quantfamdata.bed].

Calculating allele frequencies ...

Recoding genotypes (individual major mode) ...

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)

Summary of the GRM:

Mean of diagonals = 1.00124

Variance of diagonals = 0.00252032

Mean of off-diagonals = -0.00201568

Variance of off-diagonals = 0.00115717

GRM of 498 individuals has been saved in the file [output/GCTAgrmchr4.grm.bin] (in binary format).

Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrmchr4.grm.N.bin] (in binary format).

IDs for the GRM file [output/GCTAgrmchr4.grm.bin] have been saved in the file [output/GCTAgrmchr4.grm.id].

Analysis finished at 05:42:13 UTC on Thu Sep 10 2026

Overall computational time: 0.43 sec.

* Genome-wide Complex Trait Analysis (GCTA)

* version v1.95.3 Linux

* Built at Sep 9 2026 10:56:04, by GCC 12.2

* (C) 2010-present, Yang Lab, Westlake University

* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>

Analysis started at 05:42:13 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

--bfile ./data/quantfamdata

```
--chr 5
--make-grm-bin
--out output/GCTAgrmchr5
```

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

```
Reading PLINK FAM file from [./data/quantfamdata.fam].
498 individuals to be included from [./data/quantfamdata.fam].
Reading PLINK BIM file from [./data/quantfamdata.bim].
134946 SNPs to be included from [./data/quantfamdata.bim].
7263 SNPs on chromosome 5 are included in the analysis.
Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format
...
Genotype data for 498 individuals and 7263 SNPs to be included from [./data/quantfamdata.bed].
Calculating allele frequencies ...
Recoding genotypes (individual major mode) ...
```

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)

```
Summary of the GRM:
Mean of diagonals = 1.00078
Variance of diagonals = 0.00155625
Mean of off-diagonals = -0.00201473
Variance of off-diagonals = 0.00115265
GRM of 498 individuals has been saved in the file [output/GCTAgrmchr5.grm.bin] (in binary format).
Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrmchr5.grm.N.bin] (in binary format).
IDs for the GRM file [output/GCTAgrmchr5.grm.bin] have been saved in the file [output/GCTAgrmchr5.grm.id].
```

Analysis finished at 05:42:13 UTC on Thu Sep 10 2026

Overall computational time: 0.40 sec.

```
* Genome-wide Complex Trait Analysis (GCTA)
* version v1.95.3 Linux
* Built at Sep  9 2026 10:56:04, by GCC 12.2
* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
```

Analysis started at 05:42:13 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

```
--bfile ./data/quantfamdata
--chr 7
--make-grm-bin
--out output/GCTAgrmchr7
```

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

le processors in your machine.

```
Reading PLINK FAM file from [./data/quantfamdata.fam].
498 individuals to be included from [./data/quantfamdata.fam].
Reading PLINK BIM file from [./data/quantfamdata.bim].
134946 SNPs to be included from [./data/quantfamdata.bim].
6255 SNPs on chromosome 7 are included in the analysis.
Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format
...
Genotype data for 498 individuals and 6255 SNPs to be included from [./data/
quantfamdata.bed].
Calculating allele frequencies ...
Recoding genotypes (individual major mode) ...

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-o
ptimized mode, may use huge RAM)
```

```
Summary of the GRM:
Mean of diagonals = 1.00317
Variance of diagonals = 0.00207579
Mean of off-diagonals = -0.00201959
Variance of off-diagonals = 0.00119177
GRM of 498 individuals has been saved in the file [output/GCTAgrmchr7.grm.bi
n] (in binary format).
Number of SNPs to calculate the genetic relationship between each pair of in
dividuals has been saved in the file [output/GCTAgrmchr7.grm.N.bin] (in bina
ry format).
IDs for the GRM file [output/GCTAgrmchr7.grm.bin] have been saved in the fil
e [output/GCTAgrmchr7.grm.id].
```

Analysis finished at 05:42:13 UTC on Thu Sep 10 2026

Overall computational time: 0.39 sec.

```
* Genome-wide Complex Trait Analysis (GCTA)
* version v1.95.3 Linux
* Built at Sep  9 2026 10:56:04, by GCC 12.2
* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
```

Analysis started at 05:42:14 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

```
--bfile ./data/quantfamdata
--chr 8
--make-grm-bin
--out output/GCTAgrmchr8
```

Note: This is a multi-thread program. You could specify the number of thread
s by the --thread-num option to speed up the computation if there are multip
le processors in your machine.

```
Reading PLINK FAM file from [./data/quantfamdata.fam].
498 individuals to be included from [./data/quantfamdata.fam].
Reading PLINK BIM file from [./data/quantfamdata.bim].
134946 SNPs to be included from [./data/quantfamdata.bim].
```


5849 SNPs on chromosome 8 are included in the analysis.
 Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format
 ...
 Genotype data for 498 individuals and 5849 SNPs to be included from [./data/quantfamdata.bed].
 Calculating allele frequencies ...
 Recoding genotypes (individual major mode) ...

 Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)

 Summary of the GRM:
 Mean of diagonals = 1.00401
 Variance of diagonals = 0.00353961
 Mean of off-diagonals = -0.00202128
 Variance of off-diagonals = 0.00162606
 GRM of 498 individuals has been saved in the file [output/GCTAgrmchr8.grm.bin] (in binary format).
 Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrmchr8.grm.N.bin] (in binary format).
 IDs for the GRM file [output/GCTAgrmchr8.grm.bin] have been saved in the file [output/GCTAgrmchr8.grm.id].

Analysis finished at 05:42:14 UTC on Thu Sep 10 2026

Overall computational time: 0.38 sec.

* Genome-wide Complex Trait Analysis (GCTA)

* version v1.95.3 Linux

* Built at Sep 9 2026 10:56:04, by GCC 12.2

* (C) 2010-present, Yang Lab, Westlake University

* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>

Analysis started at 05:42:14 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

--bfile ./data/quantfamdata

--chr 9

--make-grm-bin

--out output/GCTAgrmchr9

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

Reading PLINK FAM file from [./data/quantfamdata.fam].

498 individuals to be included from [./data/quantfamdata.fam].

Reading PLINK BIM file from [./data/quantfamdata.bim].

134946 SNPs to be included from [./data/quantfamdata.bim].

4531 SNPs on chromosome 9 are included in the analysis.

Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format

...

Genotype data for 498 individuals and 4531 SNPs to be included from [./data/quantfamdata.bed].

Calculating allele frequencies ...

Recoding genotypes (individual major mode) ...

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)

Summary of the GRM:

Mean of diagonals = 1.00239

Variance of diagonals = 0.00213473

Mean of off-diagonals = -0.00201799

Variance of off-diagonals = 0.0013214

GRM of 498 individuals has been saved in the file [output/GCTAgrimchr9.grm.bin] (in binary format).

Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrimchr9.grm.N.bin] (in binary format).

IDs for the GRM file [output/GCTAgrimchr9.grm.bin] have been saved in the file [output/GCTAgrimchr9.grm.id].

Analysis finished at 05:42:14 UTC on Thu Sep 10 2026

Overall computational time: 0.39 sec.

* Genome-wide Complex Trait Analysis (GCTA)

* version v1.95.3 Linux

* Built at Sep 9 2026 10:56:04, by GCC 12.2

* (C) 2010-present, Yang Lab, Westlake University

* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>

Analysis started at 05:42:14 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

--bfile ./data/quantfamdata

--chr 10

--make-grm-bin

--out output/GCTAgrimchr10

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

Reading PLINK FAM file from [./data/quantfamdata.fam].

498 individuals to be included from [./data/quantfamdata.fam].

Reading PLINK BIM file from [./data/quantfamdata.bim].

134946 SNPs to be included from [./data/quantfamdata.bim].

5331 SNPs on chromosome 10 are included in the analysis.

Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format

...

Genotype data for 498 individuals and 5331 SNPs to be included from [./data/quantfamdata.bed].

Calculating allele frequencies ...

Recoding genotypes (individual major mode) ...

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)

Summary of the GRM:

Mean of diagonals = 1.00488
Variance of diagonals = 0.00260223
Mean of off-diagonals = -0.00202298
Variance of off-diagonals = 0.00137574
GRM of 498 individuals has been saved in the file [output/GCTAgrmchr10.grm.bin] (in binary format).
Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrmchr10.grm.N.bin] (in binary format).
IDs for the GRM file [output/GCTAgrmchr10.grm.bin] have been saved in the file [output/GCTAgrmchr10.grm.id].

Analysis finished at 05:42:15 UTC on Thu Sep 10 2026

Overall computational time: 0.39 sec.

* Genome-wide Complex Trait Analysis (GCTA)

* version v1.95.3 Linux

* Built at Sep 9 2026 10:56:04, by GCC 12.2

* (C) 2010-present, Yang Lab, Westlake University

* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>

Analysis started at 05:42:15 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

--bfile ./data/quantfamdata

--chr 11

--make-grm-bin

--out output/GCTAgrmchr11

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

Reading PLINK FAM file from [./data/quantfamdata.fam].

498 individuals to be included from [./data/quantfamdata.fam].

Reading PLINK BIM file from [./data/quantfamdata.bim].

134946 SNPs to be included from [./data/quantfamdata.bim].

5401 SNPs on chromosome 11 are included in the analysis.

Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format

...

Genotype data for 498 individuals and 5401 SNPs to be included from [./data/quantfamdata.bed].

Calculating allele frequencies ...

Recoding genotypes (individual major mode) ...

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)

Summary of the GRM:

Mean of diagonals = 1.00379

Variance of diagonals = 0.00331639

Mean of off-diagonals = -0.00202081

Variance of off-diagonals = 0.00152512

GRM of 498 individuals has been saved in the file [output/GCTAgrmchr11.grm.bin] (in binary format).

Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrimchr11.grm.N.bin] (in binary format).

IDs for the GRM file [output/GCTAgrimchr11.grm.bin] have been saved in the file [output/GCTAgrimchr11.grm.id].

Analysis finished at 05:42:15 UTC on Thu Sep 10 2026

Overall computational time: 0.36 sec.

* Genome-wide Complex Trait Analysis (GCTA)

* version v1.95.3 Linux

* Built at Sep 9 2026 10:56:04, by GCC 12.2

* (C) 2010-present, Yang Lab, Westlake University

* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>

Analysis started at 05:42:15 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

--bfile ./data/quantfamdata

--chr 13

--make-grm-bin

--out output/GCTAgrimchr13

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

Reading PLINK FAM file from [./data/quantfamdata.fam].

498 individuals to be included from [./data/quantfamdata.fam].

Reading PLINK BIM file from [./data/quantfamdata.bim].

134946 SNPs to be included from [./data/quantfamdata.bim].

3984 SNPs on chromosome 13 are included in the analysis.

Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format

...

Genotype data for 498 individuals and 3984 SNPs to be included from [./data/quantfamdata.bed].

Calculating allele frequencies ...

Recoding genotypes (individual major mode) ...

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)

Summary of the GRM:

Mean of diagonals = 1.00639

Variance of diagonals = 0.00387782

Mean of off-diagonals = -0.00202609

Variance of off-diagonals = 0.00165767

GRM of 498 individuals has been saved in the file [output/GCTAgrimchr13.grm.bin] (in binary format).

Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrimchr13.grm.N.bin] (in binary format).

IDs for the GRM file [output/GCTAgrimchr13.grm.bin] have been saved in the file [output/GCTAgrimchr13.grm.id].

Analysis finished at 05:42:16 UTC on Thu Sep 10 2026

Overall computational time: 0.31 sec.

* Genome-wide Complex Trait Analysis (GCTA)

* version v1.95.3 Linux

* Built at Sep 9 2026 10:56:04, by GCC 12.2

* (C) 2010-present, Yang Lab, Westlake University

* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>

Analysis started at 05:42:16 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

--bfile ./data/quantfamdata

--chr 14

--make-grm-bin

--out output/GCTAgrmchr14

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

Reading PLINK FAM file from [./data/quantfamdata.fam].

498 individuals to be included from [./data/quantfamdata.fam].

Reading PLINK BIM file from [./data/quantfamdata.bim].

134946 SNPs to be included from [./data/quantfamdata.bim].

3613 SNPs on chromosome 14 are included in the analysis.

Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format

...

Genotype data for 498 individuals and 3613 SNPs to be included from [./data/quantfamdata.bed].

Calculating allele frequencies ...

Recoding genotypes (individual major mode) ...

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)

Summary of the GRM:

Mean of diagonals = 1.00059

Variance of diagonals = 0.00343677

Mean of off-diagonals = -0.00201443

Variance of off-diagonals = 0.00176074

GRM of 498 individuals has been saved in the file [output/GCTAgrmchr14.grm.bin] (in binary format).

Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrmchr14.grm.N.bin] (in binary format).

IDs for the GRM file [output/GCTAgrmchr14.grm.bin] have been saved in the file [output/GCTAgrmchr14.grm.id].

Analysis finished at 05:42:16 UTC on Thu Sep 10 2026

Overall computational time: 0.37 sec.

* Genome-wide Complex Trait Analysis (GCTA)

* version v1.95.3 Linux

* Built at Sep 9 2026 10:56:04, by GCC 12.2

```
* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
*****
Analysis started at 05:42:16 UTC on Thu Sep 10 2026.
Hostname: 8768ca282dde
```

```
Accepted options:
--bfile ./data/quantfamdata
--chr 15
--make-grm-bin
--out output/GCTAgrmchr15
```

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

```
Reading PLINK FAM file from [./data/quantfamdata.fam].
498 individuals to be included from [./data/quantfamdata.fam].
Reading PLINK BIM file from [./data/quantfamdata.bim].
134946 SNPs to be included from [./data/quantfamdata.bim].
3215 SNPs on chromosome 15 are included in the analysis.
Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format
...
Genotype data for 498 individuals and 3215 SNPs to be included from [./data/quantfamdata.bed].
Calculating allele frequencies ...
Recoding genotypes (individual major mode) ...
```

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)

```
Summary of the GRM:
Mean of diagonals = 1.00121
Variance of diagonals = 0.00434469
Mean of off-diagonals = -0.00201567
Variance of off-diagonals = 0.00202123
GRM of 498 individuals has been saved in the file [output/GCTAgrmchr15.grm.bin] (in binary format).
Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrmchr15.grm.N.bin] (in binary format).
IDs for the GRM file [output/GCTAgrmchr15.grm.bin] have been saved in the file [output/GCTAgrmchr15.grm.id].
```

```
Analysis finished at 05:42:16 UTC on Thu Sep 10 2026
Overall computational time: 0.32 sec.
*****
* Genome-wide Complex Trait Analysis (GCTA)
* version v1.95.3 Linux
* Built at Sep 9 2026 10:56:04, by GCC 12.2
* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
*****
Analysis started at 05:42:16 UTC on Thu Sep 10 2026.
Hostname: 8768ca282dde
```

Accepted options:
--bfile ./data/quantfamdata
--chr 16
--make-grm-bin
--out output/GCTAgrmchr16

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

Reading PLINK FAM file from [./data/quantfamdata.fam].
498 individuals to be included from [./data/quantfamdata.fam].
Reading PLINK BIM file from [./data/quantfamdata.bim].
134946 SNPs to be included from [./data/quantfamdata.bim].
3050 SNPs on chromosome 16 are included in the analysis.
Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format
...
Genotype data for 498 individuals and 3050 SNPs to be included from [./data/quantfamdata.bed].
Calculating allele frequencies ...
Recoding genotypes (individual major mode) ...

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)

Summary of the GRM:
Mean of diagonals = 1.00176
Variance of diagonals = 0.00472954
Mean of off-diagonals = -0.00201675
Variance of off-diagonals = 0.0019014
GRM of 498 individuals has been saved in the file [output/GCTAgrmchr16.grm.bin] (in binary format).
Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrmchr16.grm.N.bin] (in binary format).
IDs for the GRM file [output/GCTAgrmchr16.grm.bin] have been saved in the file [output/GCTAgrmchr16.grm.id].

Analysis finished at 05:42:17 UTC on Thu Sep 10 2026

Overall computational time: 0.29 sec.

* Genome-wide Complex Trait Analysis (GCTA)
* version v1.95.3 Linux
* Built at Sep 9 2026 10:56:04, by GCC 12.2
* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>

Analysis started at 05:42:17 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:
--bfile ./data/quantfamdata
--chr 17
--make-grm-bin
--out output/GCTAgrmchr17

Note: This is a multi-thread program. You could specify the number of threads by the `--thread-num` option to speed up the computation if there are multiple processors in your machine.

```
Reading PLINK FAM file from [./data/quantfamdata.fam].
498 individuals to be included from [./data/quantfamdata.fam].
Reading PLINK BIM file from [./data/quantfamdata.bim].
134946 SNPs to be included from [./data/quantfamdata.bim].
3114 SNPs on chromosome 17 are included in the analysis.
Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format
...
Genotype data for 498 individuals and 3114 SNPs to be included from [./data/quantfamdata.bed].
Calculating allele frequencies ...
Recoding genotypes (individual major mode) ...
```

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)

```
Summary of the GRM:
Mean of diagonals = 1.00504
Variance of diagonals = 0.00443132
Mean of off-diagonals = -0.00202336
Variance of off-diagonals = 0.0020859
GRM of 498 individuals has been saved in the file [output/GCTAgrmchr17.grm.bin] (in binary format).
Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrmchr17.grm.N.bin] (in binary format).
IDs for the GRM file [output/GCTAgrmchr17.grm.bin] have been saved in the file [output/GCTAgrmchr17.grm.id].
```

Analysis finished at 05:42:17 UTC on Thu Sep 10 2026

Overall computational time: 0.29 sec.

```
* Genome-wide Complex Trait Analysis (GCTA)
* version v1.95.3 Linux
* Built at Sep  9 2026 10:56:04, by GCC 12.2
* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
```

Analysis started at 05:42:17 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

```
--bfile ./data/quantfamdata
--chr 18
--make-grm-bin
--out output/GCTAgrmchr18
```

Note: This is a multi-thread program. You could specify the number of threads by the `--thread-num` option to speed up the computation if there are multiple processors in your machine.

```
Reading PLINK FAM file from [./data/quantfamdata.fam].
498 individuals to be included from [./data/quantfamdata.fam].
```



```

Reading PLINK BIM file from [./data/quantfamdata.bim].
134946 SNPs to be included from [./data/quantfamdata.bim].
3100 SNPs on chromosome 18 are included in the analysis.
Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format
...
Genotype data for 498 individuals and 3100 SNPs to be included from [./data/
quantfamdata.bed].
Calculating allele frequencies ...
Recoding genotypes (individual major mode) ...

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-o
ptimized mode, may use huge RAM)

Summary of the GRM:
Mean of diagonals = 0.999809
Variance of diagonals = 0.00420447
Mean of off-diagonals = -0.00201278
Variance of off-diagonals = 0.00181342
GRM of 498 individuals has been saved in the file [output/GCTAgrmchr18.grm.b
in] (in binary format).
Number of SNPs to calculate the genetic relationship between each pair of in
dividuals has been saved in the file [output/GCTAgrmchr18.grm.N.bin] (in bin
ary format).
IDs for the GRM file [output/GCTAgrmchr18.grm.bin] have been saved in the fi
le [output/GCTAgrmchr18.grm.id].

```

Analysis finished at 05:42:17 UTC on Thu Sep 10 2026

Overall computational time: 0.29 sec.

```

* Genome-wide Complex Trait Analysis (GCTA)
* version v1.95.3 Linux
* Built at Sep  9 2026 10:56:04, by GCC 12.2
* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>

```

Analysis started at 05:42:17 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

```

--bfile ./data/quantfamdata
--chr 19
--make-grm-bin
--out output/GCTAgrmchr19

```

Note: This is a multi-thread program. You could specify the number of thread
s by the --thread-num option to speed up the computation if there are multip
le processors in your machine.

```

Reading PLINK FAM file from [./data/quantfamdata.fam].
498 individuals to be included from [./data/quantfamdata.fam].
Reading PLINK BIM file from [./data/quantfamdata.bim].
134946 SNPs to be included from [./data/quantfamdata.bim].
2288 SNPs on chromosome 19 are included in the analysis.
Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format
...
Genotype data for 498 individuals and 2288 SNPs to be included from [./data/

```

```
quantfamdata.bed].
Calculating allele frequencies ...
Recoding genotypes (individual major mode) ...
```

```
Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)
```

```
Summary of the GRM:
```

```
Mean of diagonals = 1.00217
```

```
Variance of diagonals = 0.00398606
```

```
Mean of off-diagonals = -0.0020176
```

```
Variance of off-diagonals = 0.00212117
```

```
GRM of 498 individuals has been saved in the file [output/GCTAgrmchr19.grm.bin] (in binary format).
```

```
Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrmchr19.grm.N.bin] (in binary format).
```

```
IDs for the GRM file [output/GCTAgrmchr19.grm.bin] have been saved in the file [output/GCTAgrmchr19.grm.id].
```

```
Analysis finished at 05:42:18 UTC on Thu Sep 10 2026
```

```
Overall computational time: 0.27 sec.
```

```
*****
```

```
* Genome-wide Complex Trait Analysis (GCTA)
```

```
* version v1.95.3 Linux
```

```
* Built at Sep 9 2026 10:56:04, by GCC 12.2
```

```
* (C) 2010-present, Yang Lab, Westlake University
```

```
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
```

```
*****
```

```
Analysis started at 05:42:18 UTC on Thu Sep 10 2026.
```

```
Hostname: 8768ca282dde
```

```
Accepted options:
```

```
--bfile ./data/quantfamdata
```

```
--chr 20
```

```
--make-grm-bin
```

```
--out output/GCTAgrmchr20
```

```
Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.
```

```
Reading PLINK FAM file from [./data/quantfamdata.fam].
```

```
498 individuals to be included from [./data/quantfamdata.fam].
```

```
Reading PLINK BIM file from [./data/quantfamdata.bim].
```

```
134946 SNPs to be included from [./data/quantfamdata.bim].
```

```
2472 SNPs on chromosome 20 are included in the analysis.
```

```
Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format
```

```
...
```

```
Genotype data for 498 individuals and 2472 SNPs to be included from [./data/quantfamdata.bed].
```

```
Calculating allele frequencies ...
```

```
Recoding genotypes (individual major mode) ...
```

```
Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)
```

Summary of the GRM:

Mean of diagonals = 1.00077

Variance of diagonals = 0.00422422

Mean of off-diagonals = -0.00201468

Variance of off-diagonals = 0.00198072

GRM of 498 individuals has been saved in the file [output/GCTAgrmchr20.grm.bin] (in binary format).

Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrmchr20.grm.N.bin] (in binary format).

IDs for the GRM file [output/GCTAgrmchr20.grm.bin] have been saved in the file [output/GCTAgrmchr20.grm.id].

Analysis finished at 05:42:18 UTC on Thu Sep 10 2026

Overall computational time: 0.29 sec.

* Genome-wide Complex Trait Analysis (GCTA)

* version v1.95.3 Linux

* Built at Sep 9 2026 10:56:04, by GCC 12.2

* (C) 2010-present, Yang Lab, Westlake University

* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>

Analysis started at 05:42:18 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

--bfile ./data/quantfamdata

--chr 21

--make-grm-bin

--out output/GCTAgrmchr21

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

Reading PLINK FAM file from [./data/quantfamdata.fam].

498 individuals to be included from [./data/quantfamdata.fam].

Reading PLINK BIM file from [./data/quantfamdata.bim].

134946 SNPs to be included from [./data/quantfamdata.bim].

1402 SNPs on chromosome 21 are included in the analysis.

Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format

...

Genotype data for 498 individuals and 1402 SNPs to be included from [./data/quantfamdata.bed].

Calculating allele frequencies ...

Recoding genotypes (individual major mode) ...

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)

Summary of the GRM:

Mean of diagonals = 1.00425

Variance of diagonals = 0.00596999

Mean of off-diagonals = -0.00202175

Variance of off-diagonals = 0.00282492

GRM of 498 individuals has been saved in the file [output/GCTAgrimchr21.grm.bin] (in binary format).
Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrimchr21.grm.N.bin] (in binary format).
IDs for the GRM file [output/GCTAgrimchr21.grm.bin] have been saved in the file [output/GCTAgrimchr21.grm.id].

Analysis finished at 05:42:18 UTC on Thu Sep 10 2026

Overall computational time: 0.28 sec.

* Genome-wide Complex Trait Analysis (GCTA)

* version v1.95.3 Linux

* Built at Sep 9 2026 10:56:04, by GCC 12.2

* (C) 2010-present, Yang Lab, Westlake University

* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>

Analysis started at 05:42:18 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

--bfile ./data/quantfamdata

--chr 22

--make-grm-bin

--out output/GCTAgrimchr22

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

Reading PLINK FAM file from [./data/quantfamdata.fam].

498 individuals to be included from [./data/quantfamdata.fam].

Reading PLINK BIM file from [./data/quantfamdata.bim].

134946 SNPs to be included from [./data/quantfamdata.bim].

1384 SNPs on chromosome 22 are included in the analysis.

Reading PLINK BED file from [./data/quantfamdata.bed] in SNP-major format

...

Genotype data for 498 individuals and 1384 SNPs to be included from [./data/quantfamdata.bed].

Calculating allele frequencies ...

Recoding genotypes (individual major mode) ...

Calculating the genetic relationship matrix (GRM) ... (Note: default speed-optimized mode, may use huge RAM)

Summary of the GRM:

Mean of diagonals = 1.0008

Variance of diagonals = 0.00553361

Mean of off-diagonals = -0.00201503

Variance of off-diagonals = 0.00287841

GRM of 498 individuals has been saved in the file [output/GCTAgrimchr22.grm.bin] (in binary format).

Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrimchr22.grm.N.bin] (in binary format).

IDs for the GRM file [output/GCTAgrimchr22.grm.bin] have been saved in the fi

```
le [output/GCTAgrmchr22.grm.id].
```

Analysis finished at 05:42:19 UTC on Thu Sep 10 2026
Overall computational time: 0.28 sec.

Then run REML with all 22 chromosome-specific GRMs simultaneously (`data/multipleGRMs.txt` lists their stem names — take a look at it). In this dataset the analysis fails to converge, likely because this kind of full-partition analysis really needs a larger set of less closely related individuals than we have here.

As a more tractable alternative, we partition heritability into just two sets — chromosome 6 versus everything else. First combine the non-chr6 GRMs into one (`data/multipleGRMsnot6.txt` lists their stems), then run REML with both pieces (`data/multipleGRMs6andnot6.txt` lists chr6 and the combined rest). Expect a total heritability of about 0.47, split roughly 0.29 for chromosome 6 and 0.17 for the remaining autosomes.

```
In [7]: # Attempt REML with all 22 per-chromosome GRMs at once (expected not to conv
gcta64 --reml --mgrm-bin ./data/multipleGRMs.txt --pheno ./data/myphenos.txt

# Combine the 21 non-chr6 GRMs into a single merged GRM
gcta64 --mgrm-bin ./data/multipleGRMsnot6.txt --make-grm --out output/GCTAgr

# REML with just two variance components: chr6 alone, and the combined rest
gcta64 --reml --mgrm-bin ./data/multipleGRMs6andnot6.txt --pheno ./data/myph
--out output/GCTAherit6andnot6
```

```
*****
* Genome-wide Complex Trait Analysis (GCTA)
* version v1.95.3 Linux
* Built at Sep  9 2026 10:56:04, by GCC 12.2
* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
*****
Analysis started at 05:42:19 UTC on Thu Sep 10 2026.
Hostname: 8768ca282dde
```

Accepted options:

```
--reml
--mgrm-bin ./data/multipleGRMs.txt
--pheno ./data/myphenos.txt
--out output/GCTAherit22GRMs
```

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

Reading phenotypes from [./data/myphenos.txt].

Non-missing phenotypes of 498 individuals are included from [./data/myphenos.txt].

There are 22 GRM file names specified in the file [./data/multipleGRMs.txt].

Reading the GRM from the 1th file ...

Reading IDs of the GRM from [output/GCTAgrmchr1.grm.id].

498 IDs are read from [output/GCTAgrmchr1.grm.id].

Reading the GRM from [output/GCTAgrmchr1.grm.bin].

GRM for 498 individuals are included from [output/GCTAgrmchr1.grm.bin].

Reading the GRM from the 2th file ...

Reading IDs of the GRM from [output/GCTAgrmchr2.grm.id].

498 IDs are read from [output/GCTAgrmchr2.grm.id].

Reading the GRM from [output/GCTAgrmchr2.grm.bin].

GRM for 498 individuals are included from [output/GCTAgrmchr2.grm.bin].

Reading the GRM from the 3th file ...

Reading IDs of the GRM from [output/GCTAgrmchr3.grm.id].

498 IDs are read from [output/GCTAgrmchr3.grm.id].

Reading the GRM from [output/GCTAgrmchr3.grm.bin].

GRM for 498 individuals are included from [output/GCTAgrmchr3.grm.bin].

Reading the GRM from the 4th file ...

Reading IDs of the GRM from [output/GCTAgrmchr4.grm.id].

498 IDs are read from [output/GCTAgrmchr4.grm.id].

Reading the GRM from [output/GCTAgrmchr4.grm.bin].

GRM for 498 individuals are included from [output/GCTAgrmchr4.grm.bin].

Reading the GRM from the 5th file ...

Reading IDs of the GRM from [output/GCTAgrmchr5.grm.id].

498 IDs are read from [output/GCTAgrmchr5.grm.id].

Reading the GRM from [output/GCTAgrmchr5.grm.bin].

GRM for 498 individuals are included from [output/GCTAgrmchr5.grm.bin].

Reading the GRM from the 6th file ...

Reading IDs of the GRM from [output/GCTAgrmchr6.grm.id].

498 IDs are read from [output/GCTAgrmchr6.grm.id].

Reading the GRM from [output/GCTAgrmchr6.grm.bin].

GRM for 498 individuals are included from [output/GCTAgrmchr6.grm.bin].

Reading the GRM from the 7th file ...

Reading IDs of the GRM from [output/GCTAgrmchr7.grm.id].
498 IDs are read from [output/GCTAgrmchr7.grm.id].
Reading the GRM from [output/GCTAgrmchr7.grm.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr7.grm.bin].
Reading the GRM from the 8th file ...
Reading IDs of the GRM from [output/GCTAgrmchr8.grm.id].
498 IDs are read from [output/GCTAgrmchr8.grm.id].
Reading the GRM from [output/GCTAgrmchr8.grm.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr8.grm.bin].
Reading the GRM from the 9th file ...
Reading IDs of the GRM from [output/GCTAgrmchr9.grm.id].
498 IDs are read from [output/GCTAgrmchr9.grm.id].
Reading the GRM from [output/GCTAgrmchr9.grm.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr9.grm.bin].
Reading the GRM from the 10th file ...
Reading IDs of the GRM from [output/GCTAgrmchr10.grm.id].
498 IDs are read from [output/GCTAgrmchr10.grm.id].
Reading the GRM from [output/GCTAgrmchr10.grm.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr10.grm.bin].
Reading the GRM from the 11th file ...
Reading IDs of the GRM from [output/GCTAgrmchr11.grm.id].
498 IDs are read from [output/GCTAgrmchr11.grm.id].
Reading the GRM from [output/GCTAgrmchr11.grm.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr11.grm.bin].
Reading the GRM from the 12th file ...
Reading IDs of the GRM from [output/GCTAgrmchr12.grm.id].
498 IDs are read from [output/GCTAgrmchr12.grm.id].
Reading the GRM from [output/GCTAgrmchr12.grm.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr12.grm.bin].
Reading the GRM from the 13th file ...
Reading IDs of the GRM from [output/GCTAgrmchr13.grm.id].
498 IDs are read from [output/GCTAgrmchr13.grm.id].
Reading the GRM from [output/GCTAgrmchr13.grm.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr13.grm.bin].
Reading the GRM from the 14th file ...
Reading IDs of the GRM from [output/GCTAgrmchr14.grm.id].
498 IDs are read from [output/GCTAgrmchr14.grm.id].
Reading the GRM from [output/GCTAgrmchr14.grm.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr14.grm.bin].
Reading the GRM from the 15th file ...
Reading IDs of the GRM from [output/GCTAgrmchr15.grm.id].
498 IDs are read from [output/GCTAgrmchr15.grm.id].
Reading the GRM from [output/GCTAgrmchr15.grm.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr15.grm.bin].
Reading the GRM from the 16th file ...
Reading IDs of the GRM from [output/GCTAgrmchr16.grm.id].
498 IDs are read from [output/GCTAgrmchr16.grm.id].
Reading the GRM from [output/GCTAgrmchr16.grm.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr16.grm.bin].
Reading the GRM from the 17th file ...
Reading IDs of the GRM from [output/GCTAgrmchr17.grm.id].
498 IDs are read from [output/GCTAgrmchr17.grm.id].
Reading the GRM from [output/GCTAgrmchr17.grm.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr17.grm.bin].
Reading the GRM from the 18th file ...
Reading IDs of the GRM from [output/GCTAgrmchr18.grm.id].

945	0.00006	1.51995	0.00006	0.92639	2.78834	1.95803	35.07728	(7 c
omponent(s) constrained)								
6	-1275.80		0.22101	0.00006	2.95269	0.00006	0.15053	17.77251
0.52178	1.23469	0.00006	0.00006	2.72644	7.32412	3.50996	0.00006	5.60771 3.37
864	0.00006	1.22454	0.00006	0.87657	3.01691	2.16861	35.18062	(7 c
omponent(s) constrained)								
7	-1280.74		0.28784	0.00006	3.04144	0.00006	0.27020	18.20071
0.63898	1.37742	0.00006	0.00006	2.69250	7.11866	0.81318	0.00006	5.58635 0.99
569	0.00006	1.37753	0.00006	0.85585	3.04801	2.11362	35.94228	(7 c
omponent(s) constrained)								
8	-1277.70		0.12970	0.00006	2.66705	0.00006	0.22085	17.94282
0.54267	1.10824	0.00006	0.00006	2.71635	7.29459	3.79416	0.00006	5.51139 3.62
114	0.00006	1.37267	0.00006	0.99427	2.94600	2.16952	35.30478	(7 c
omponent(s) constrained)								
9	-1281.18		0.26123	0.00006	2.95479	0.00006	0.28853	18.30237
0.65871	1.35454	0.00006	0.00006	2.70078	7.11081	0.84880	0.00006	5.55551 1.05
870	0.00006	1.43278	0.00006	0.88634	3.02170	2.10124	36.05284	(7 c
omponent(s) constrained)								
10	-1277.89		0.15539	0.00006	2.75320	0.00006	0.19546	17.89422
0.51879	1.14812	0.00006	0.00006	2.71892	7.31066	3.76856	0.00006	5.54418 3.58
692	0.00006	1.32710	0.00006	0.96010	2.96668	2.16866	35.29636	(7 c
omponent(s) constrained)								
11	-1281.15		0.26980	0.00006	2.98515	0.00006	0.28340	18.28152
0.65093	1.36617	0.00006	0.00006	2.69820	7.11514	0.84740	0.00006	5.56801 1.04
798	0.00006	1.41752	0.00006	0.87678	3.03022	2.10343	36.04732	(7 c
omponent(s) constrained)								
12	-1277.89		0.14742	0.00006	2.72635	0.00006	0.20424	17.91216
0.52846	1.13493	0.00006	0.00006	2.71865	7.30496	3.77405	0.00006	5.53458 3.59
636	0.00006	1.34218	0.00006	0.97144	2.96089	2.16953	35.30043	(7 c
omponent(s) constrained)								
13	-1281.16		0.26713	0.00006	2.97582	0.00006	0.28521	18.28950
0.65383	1.36210	0.00006	0.00006	2.69934	7.11364	0.84695	0.00006	5.56453 1.05
069	0.00006	1.42260	0.00006	0.88011	3.02800	2.10314	36.05031	(7 c
omponent(s) constrained)								
14	-1277.89		0.14987	0.00006	2.73481	0.00006	0.20120	17.90594
0.52489	1.13918	0.00006	0.00006	2.71869	7.30722	3.77336	0.00006	5.53755 3.59
421	0.00006	1.33711	0.00006	0.96773	2.96261	2.16920	35.29908	(7 c
omponent(s) constrained)								
15	-1281.16		0.26797	0.00006	2.97877	0.00006	0.28458	18.28680
0.65283	1.36348	0.00006	0.00006	2.69894	7.11429	0.84731	0.00006	5.56559 1.05
010	0.00006	1.42090	0.00006	0.87898	3.02865	2.10317	36.04942	(7 c
omponent(s) constrained)								
16	-1277.89		0.14912	0.00006	2.73210	0.00006	0.20224	17.90808
0.52612	1.13781	0.00006	0.00006	2.71868	7.30635	3.77339	0.00006	5.53661 3.59
470	0.00006	1.33883	0.00006	0.96895	2.96209	2.16932	35.29958	(7 c
omponent(s) constrained)								
17	-1281.16		0.26771	0.00006	2.97783	0.00006	0.28480	18.28772
0.65316	1.36303	0.00006	0.00006	2.69907	7.11402	0.84715	0.00006	5.56526 1.05
022	0.00006	1.42148	0.00006	0.87936	3.02845	2.10317	36.04973	(7 c
omponent(s) constrained)								
18	-1277.89		0.14935	0.00006	2.73298	0.00006	0.20189	17.90735
0.52571	1.13825	0.00006	0.00006	2.71868	7.30668	3.77342	0.00006	5.53692 3.59
460	0.00006	1.33824	0.00006	0.96855	2.96225	2.16928	35.29939	(7 c
omponent(s) constrained)								
19	-1281.16		0.26779	0.00006	2.97813	0.00006	0.28472	18.28741
0.65305	1.36318	0.00006	0.00006	2.69903	7.11413	0.84721	0.00006	5.56536 1.05

020	0.00006	1.42128	0.00006	0.87924	3.02851	2.10317	36.04962	(7 c
omponent(s) constrained)								
20	-1277.89		0.14928	0.00006	2.73270	0.00006	0.20201	17.90760
0.52585	1.13811	0.00006	0.00006	2.71868	7.30656	3.77340	0.00006	5.53682 3.59
461	0.00006	1.33844	0.00006	0.96868	2.96220	2.16929	35.29946	(7 c
omponent(s) constrained)								
21	-1281.16		0.26776	0.00006	2.97803	0.00006	0.28475	18.28751
0.65309	1.36313	0.00006	0.00006	2.69904	7.11409	0.84719	0.00006	5.56533 1.05
020	0.00006	1.42135	0.00006	0.87928	3.02849	2.10317	36.04966	(7 c
omponent(s) constrained)								
22	-1277.89		0.14930	0.00006	2.73279	0.00006	0.20197	17.90752
0.52580	1.13815	0.00006	0.00006	2.71868	7.30660	3.77341	0.00006	5.53685 3.59
461	0.00006	1.33837	0.00006	0.96864	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
23	-1281.16		0.26777	0.00006	2.97807	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534 1.05
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04964	(7 c
omponent(s) constrained)								
24	-1277.89		0.14929	0.00006	2.73276	0.00006	0.20198	17.90755
0.52582	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684 3.59
461	0.00006	1.33840	0.00006	0.96865	2.96221	2.16929	35.29945	(7 c
omponent(s) constrained)								
25	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28749
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534 1.05
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
26	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684 3.59
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
27	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534 1.05
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
28	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684 3.59
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
29	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534 1.05
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
30	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684 3.59
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
31	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534 1.05
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
32	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684 3.59
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
33	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534 1.05

020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
34	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
35	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
36	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
37	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
38	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
39	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
40	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
41	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
42	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
43	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
44	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
45	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7

020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
48	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
49	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
50	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
51	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
52	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
53	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
54	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
55	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
56	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
57	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
58	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
59	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7

020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965		(7 c
omponent(s) constrained)									
62	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754	
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684	3.59
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944		(7 c
omponent(s) constrained)									
63	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748	
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534	1.05
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965		(7 c
omponent(s) constrained)									
64	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754	
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684	3.59
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944		(7 c
omponent(s) constrained)									
65	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748	
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534	1.05
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965		(7 c
omponent(s) constrained)									
66	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754	
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684	3.59
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944		(7 c
omponent(s) constrained)									
67	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748	
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534	1.05
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965		(7 c
omponent(s) constrained)									
68	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754	
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684	3.59
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944		(7 c
omponent(s) constrained)									
69	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748	
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534	1.05
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965		(7 c
omponent(s) constrained)									
70	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754	
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684	3.59
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944		(7 c
omponent(s) constrained)									
71	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748	
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534	1.05
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965		(7 c
omponent(s) constrained)									
72	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754	
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684	3.59
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944		(7 c
omponent(s) constrained)									
73	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748	

020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
76	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
77	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
78	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
79	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
80	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
81	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
82	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
83	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
84	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
85	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7 c
omponent(s) constrained)								
86	-1277.89		0.14929	0.00006	2.73277	0.00006	0.20198	17.90754
0.52581	1.13814	0.00006	0.00006	2.71868	7.30659	3.77341	0.00006	5.53684
461	0.00006	1.33839	0.00006	0.96865	2.96221	2.16929	35.29944	(7 c
omponent(s) constrained)								
87	-1281.16		0.26777	0.00006	2.97806	0.00006	0.28474	18.28748
0.65308	1.36314	0.00006	0.00006	2.69904	7.11410	0.84720	0.00006	5.56534
020	0.00006	1.42133	0.00006	0.87927	3.02850	2.10317	36.04965	(7

```

020      0.00006 1.42133 0.00006 0.87927 3.02850 2.10317 36.04965      (7 c
omponent(s) constrained)
90      -1277.89      0.14929 0.00006 2.73277 0.00006 0.20198 17.90754
0.52581 1.13814 0.00006 0.00006 2.71868 7.30659 3.77341 0.00006 5.53684 3.59
461      0.00006 1.33839 0.00006 0.96865 2.96221 2.16929 35.29944      (7 c
omponent(s) constrained)
91      -1281.16      0.26777 0.00006 2.97806 0.00006 0.28474 18.28748
0.65308 1.36314 0.00006 0.00006 2.69904 7.11410 0.84720 0.00006 5.56534 1.05
020      0.00006 1.42133 0.00006 0.87927 3.02850 2.10317 36.04965      (7 c
omponent(s) constrained)
92      -1277.89      0.14929 0.00006 2.73277 0.00006 0.20198 17.90754
0.52581 1.13814 0.00006 0.00006 2.71868 7.30659 3.77341 0.00006 5.53684 3.59
461      0.00006 1.33839 0.00006 0.96865 2.96221 2.16929 35.29944      (7 c
omponent(s) constrained)
93      -1281.16      0.26777 0.00006 2.97806 0.00006 0.28474 18.28748
0.65308 1.36314 0.00006 0.00006 2.69904 7.11410 0.84720 0.00006 5.56534 1.05
020      0.00006 1.42133 0.00006 0.87927 3.02850 2.10317 36.04965      (7 c
omponent(s) constrained)
94      -1277.89      0.14929 0.00006 2.73277 0.00006 0.20198 17.90754
0.52581 1.13814 0.00006 0.00006 2.71868 7.30659 3.77341 0.00006 5.53684 3.59
461      0.00006 1.33839 0.00006 0.96865 2.96221 2.16929 35.29944      (7 c
omponent(s) constrained)
95      -1281.16      0.26777 0.00006 2.97806 0.00006 0.28474 18.28748
0.65308 1.36314 0.00006 0.00006 2.69904 7.11410 0.84720 0.00006 5.56534 1.05
020      0.00006 1.42133 0.00006 0.87927 3.02850 2.10317 36.04965      (7 c
omponent(s) constrained)
96      -1277.89      0.14929 0.00006 2.73277 0.00006 0.20198 17.90754
0.52581 1.13814 0.00006 0.00006 2.71868 7.30659 3.77341 0.00006 5.53684 3.59
461      0.00006 1.33839 0.00006 0.96865 2.96221 2.16929 35.29944      (7 c
omponent(s) constrained)
97      -1281.16      0.26777 0.00006 2.97806 0.00006 0.28474 18.28748
0.65308 1.36314 0.00006 0.00006 2.69904 7.11410 0.84720 0.00006 5.56534 1.05
020      0.00006 1.42133 0.00006 0.87927 3.02850 2.10317 36.04965      (7 c
omponent(s) constrained)
98      -1277.89      0.14929 0.00006 2.73277 0.00006 0.20198 17.90754
0.52581 1.13814 0.00006 0.00006 2.71868 7.30659 3.77341 0.00006 5.53684 3.59
461      0.00006 1.33839 0.00006 0.96865 2.96221 2.16929 35.29944      (7 c
omponent(s) constrained)
99      -1281.16      0.26777 0.00006 2.97806 0.00006 0.28474 18.28748
0.65308 1.36314 0.00006 0.00006 2.69904 7.11410 0.84720 0.00006 5.56534 1.05
020      0.00006 1.42133 0.00006 0.87927 3.02850 2.10317 36.04965      (7 c
omponent(s) constrained)

```

Error: Log-likelihood not converged (stop after 100 interactions).
You can specify the option `--reml-maxit` to allow for more iterations.

An error occurs, please check the options or data

```

*****
* Genome-wide Complex Trait Analysis (GCTA)
* version v1.95.3 Linux
* Built at Sep  9 2026 10:56:04, by GCC 12.2
* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
*****

```

Analysis started at 05:42:23 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

--mgrm-bin ./data/multipleGRMsnot6.txt

--make-grm

--out output/GCTAgrmnot6

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

There are 21 GRM file names specified in [./data/multipleGRMsnot6.txt].

498 individuals in common in the GRM files.

Reading the GRM from the 1th file ...

Reading IDs of the GRM from [output/GCTAgrmchr1.grm.id].

498 IDs are read from [output/GCTAgrmchr1.grm.id].

Reading the GRM from [output/GCTAgrmchr1.grm.bin].

Reading the number of SNPs for the GRM from [output/GCTAgrmchr1.grm.N.bin].

GRM for 498 individuals are included from [output/GCTAgrmchr1.grm.bin].

Reading the GRM from the 2th file ...

Reading IDs of the GRM from [output/GCTAgrmchr2.grm.id].

498 IDs are read from [output/GCTAgrmchr2.grm.id].

Reading the GRM from [output/GCTAgrmchr2.grm.bin].

Reading the number of SNPs for the GRM from [output/GCTAgrmchr2.grm.N.bin].

GRM for 498 individuals are included from [output/GCTAgrmchr2.grm.bin].

Reading the GRM from the 3th file ...

Reading IDs of the GRM from [output/GCTAgrmchr3.grm.id].

498 IDs are read from [output/GCTAgrmchr3.grm.id].

Reading the GRM from [output/GCTAgrmchr3.grm.bin].

Reading the number of SNPs for the GRM from [output/GCTAgrmchr3.grm.N.bin].

GRM for 498 individuals are included from [output/GCTAgrmchr3.grm.bin].

Reading the GRM from the 4th file ...

Reading IDs of the GRM from [output/GCTAgrmchr4.grm.id].

498 IDs are read from [output/GCTAgrmchr4.grm.id].

Reading the GRM from [output/GCTAgrmchr4.grm.bin].

Reading the number of SNPs for the GRM from [output/GCTAgrmchr4.grm.N.bin].

GRM for 498 individuals are included from [output/GCTAgrmchr4.grm.bin].

Reading the GRM from the 5th file ...

Reading IDs of the GRM from [output/GCTAgrmchr5.grm.id].

498 IDs are read from [output/GCTAgrmchr5.grm.id].

Reading the GRM from [output/GCTAgrmchr5.grm.bin].

Reading the number of SNPs for the GRM from [output/GCTAgrmchr5.grm.N.bin].

GRM for 498 individuals are included from [output/GCTAgrmchr5.grm.bin].

Reading the GRM from the 6th file ...

Reading IDs of the GRM from [output/GCTAgrmchr7.grm.id].

498 IDs are read from [output/GCTAgrmchr7.grm.id].

Reading the GRM from [output/GCTAgrmchr7.grm.bin].

Reading the number of SNPs for the GRM from [output/GCTAgrmchr7.grm.N.bin].

GRM for 498 individuals are included from [output/GCTAgrmchr7.grm.bin].

Reading the GRM from the 7th file ...

Reading IDs of the GRM from [output/GCTAgrmchr8.grm.id].

498 IDs are read from [output/GCTAgrmchr8.grm.id].

Reading the GRM from [output/GCTAgrmchr8.grm.bin].

Reading the number of SNPs for the GRM from [output/GCTAgrmchr8.grm.N.bin].

GRM for 498 individuals are included from [output/GCTAgrmchr8.grm.bin].

Reading the GRM from the 8th file ...

Reading IDs of the GRM from [output/GCTAgrmchr9.grm.id].

498 IDs are read from [output/GCTAgrmchr9.grm.id].


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Reading the GRM from [output/GCTAgrmchr9.grm.bin].
Reading the number of SNPs for the GRM from [output/GCTAgrmchr9.grm.N.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr9.grm.bin].
Reading the GRM from the 9th file ...
Reading IDs of the GRM from [output/GCTAgrmchr10.grm.id].
498 IDs are read from [output/GCTAgrmchr10.grm.id].
Reading the GRM from [output/GCTAgrmchr10.grm.bin].
Reading the number of SNPs for the GRM from [output/GCTAgrmchr10.grm.N.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr10.grm.bin].
Reading the GRM from the 10th file ...
Reading IDs of the GRM from [output/GCTAgrmchr11.grm.id].
498 IDs are read from [output/GCTAgrmchr11.grm.id].
Reading the GRM from [output/GCTAgrmchr11.grm.bin].
Reading the number of SNPs for the GRM from [output/GCTAgrmchr11.grm.N.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr11.grm.bin].
Reading the GRM from the 11th file ...
Reading IDs of the GRM from [output/GCTAgrmchr12.grm.id].
498 IDs are read from [output/GCTAgrmchr12.grm.id].
Reading the GRM from [output/GCTAgrmchr12.grm.bin].
Reading the number of SNPs for the GRM from [output/GCTAgrmchr12.grm.N.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr12.grm.bin].
Reading the GRM from the 12th file ...
Reading IDs of the GRM from [output/GCTAgrmchr13.grm.id].
498 IDs are read from [output/GCTAgrmchr13.grm.id].
Reading the GRM from [output/GCTAgrmchr13.grm.bin].
Reading the number of SNPs for the GRM from [output/GCTAgrmchr13.grm.N.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr13.grm.bin].
Reading the GRM from the 13th file ...
Reading IDs of the GRM from [output/GCTAgrmchr14.grm.id].
498 IDs are read from [output/GCTAgrmchr14.grm.id].
Reading the GRM from [output/GCTAgrmchr14.grm.bin].
Reading the number of SNPs for the GRM from [output/GCTAgrmchr14.grm.N.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr14.grm.bin].
Reading the GRM from the 14th file ...
Reading IDs of the GRM from [output/GCTAgrmchr15.grm.id].
498 IDs are read from [output/GCTAgrmchr15.grm.id].
Reading the GRM from [output/GCTAgrmchr15.grm.bin].
Reading the number of SNPs for the GRM from [output/GCTAgrmchr15.grm.N.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr15.grm.bin].
Reading the GRM from the 15th file ...
Reading IDs of the GRM from [output/GCTAgrmchr16.grm.id].
498 IDs are read from [output/GCTAgrmchr16.grm.id].
Reading the GRM from [output/GCTAgrmchr16.grm.bin].
Reading the number of SNPs for the GRM from [output/GCTAgrmchr16.grm.N.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr16.grm.bin].
Reading the GRM from the 16th file ...
Reading IDs of the GRM from [output/GCTAgrmchr17.grm.id].
498 IDs are read from [output/GCTAgrmchr17.grm.id].
Reading the GRM from [output/GCTAgrmchr17.grm.bin].
Reading the number of SNPs for the GRM from [output/GCTAgrmchr17.grm.N.bin].
GRM for 498 individuals are included from [output/GCTAgrmchr17.grm.bin].
Reading the GRM from the 17th file ...
Reading IDs of the GRM from [output/GCTAgrmchr18.grm.id].
498 IDs are read from [output/GCTAgrmchr18.grm.id].
Reading the GRM from [output/GCTAgrmchr18.grm.bin].
Reading the number of SNPs for the GRM from [output/GCTAgrmchr18.grm.N.bin]

```

GRM for 498 individuals are included from [output/GCTAgrimchr18.grm.bin].
 Reading the GRM from the 18th file ...
 Reading IDs of the GRM from [output/GCTAgrimchr19.grm.id].
 498 IDs are read from [output/GCTAgrimchr19.grm.id].
 Reading the GRM from [output/GCTAgrimchr19.grm.bin].
 Reading the number of SNPs for the GRM from [output/GCTAgrimchr19.grm.N.bin].
 GRM for 498 individuals are included from [output/GCTAgrimchr19.grm.bin].
 Reading the GRM from the 19th file ...
 Reading IDs of the GRM from [output/GCTAgrimchr20.grm.id].
 498 IDs are read from [output/GCTAgrimchr20.grm.id].
 Reading the GRM from [output/GCTAgrimchr20.grm.bin].
 Reading the number of SNPs for the GRM from [output/GCTAgrimchr20.grm.N.bin].
 GRM for 498 individuals are included from [output/GCTAgrimchr20.grm.bin].
 Reading the GRM from the 20th file ...
 Reading IDs of the GRM from [output/GCTAgrimchr21.grm.id].
 498 IDs are read from [output/GCTAgrimchr21.grm.id].
 Reading the GRM from [output/GCTAgrimchr21.grm.bin].
 Reading the number of SNPs for the GRM from [output/GCTAgrimchr21.grm.N.bin].
 GRM for 498 individuals are included from [output/GCTAgrimchr21.grm.bin].
 Reading the GRM from the 21th file ...
 Reading IDs of the GRM from [output/GCTAgrimchr22.grm.id].
 498 IDs are read from [output/GCTAgrimchr22.grm.id].
 Reading the GRM from [output/GCTAgrimchr22.grm.bin].
 Reading the number of SNPs for the GRM from [output/GCTAgrimchr22.grm.N.bin].
 GRM for 498 individuals are included from [output/GCTAgrimchr22.grm.bin].

21 GRMs have been merged together.
 GRM of 498 individuals has been saved in the file [output/GCTAgrimnot6.grm.bin] (in binary format).
 Number of SNPs to calculate the genetic relationship between each pair of individuals has been saved in the file [output/GCTAgrimnot6.grm.N.bin] (in binary format).
 IDs for the GRM file [output/GCTAgrimnot6.grm.bin] have been saved in the file [output/GCTAgrimnot6.grm.id].

Analysis finished at 05:42:23 UTC on Thu Sep 10 2026

Overall computational time: 0.09 sec.

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* Built at Sep 9 2026 10:56:04, by GCC 12.2

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* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>

Analysis started at 05:42:23 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Accepted options:

--reml

--mgrm-bin ./data/multipleGRMs6andnot6.txt

--pheno ./data/myphenos.txt

--out output/GCTAherit6andnot6

Note: This is a multi-thread program. You could specify the number of threads by the --thread-num option to speed up the computation if there are multiple processors in your machine.

Reading phenotypes from [./data/myphenos.txt].

Non-missing phenotypes of 498 individuals are included from [./data/myphenos.txt].

There are 2 GRM file names specified in the file [./data/multipleGRMs6andnot6.txt].

Reading the GRM from the 1th file ...

Reading IDs of the GRM from [output/GCTAgrmchr6.grm.id].

498 IDs are read from [output/GCTAgrmchr6.grm.id].

Reading the GRM from [output/GCTAgrmchr6.grm.bin].

GRM for 498 individuals are included from [output/GCTAgrmchr6.grm.bin].

Reading the GRM from the 2th file ...

Reading IDs of the GRM from [output/GCTAgrmnot6.grm.id].

498 IDs are read from [output/GCTAgrmnot6.grm.id].

Reading the GRM from [output/GCTAgrmnot6.grm.bin].

GRM for 498 individuals are included from [output/GCTAgrmnot6.grm.bin].

498 individuals are in common in these files.

Performing REML analysis ... (Note: may take hours depending on sample size).

498 observations, 1 fixed effect(s), and 3 variance component(s)(including residual variance).

Calculating prior values of variance components by EM-REML ...

Updated prior values: 20.748 20.7874 21.1358

logL: -1270.78

Running AI-REML algorithm ...

Iter.	logL	V(G1)	V(G2)	V(e)
1	-1270.71	19.93932	17.28038	24.98948
2	-1270.08	18.14765	9.98903	33.38078
3	-1269.60	18.18990	10.90080	32.73385
4	-1269.59	18.20629	10.78833	32.82773
5	-1269.59	18.20278	10.80439	32.81581
6	-1269.59	18.20343	10.80210	32.81739

Log-likelihood ratio converged.

Calculating the logLikelihood for the reduced model ...

(variance component 1 is dropped from the model)

Calculating prior values of variance components by EM-REML ...

Updated prior values: 31.04062 31.40201

logL: -1274.35341

Running AI-REML algorithm ...

Iter.	logL	V(G2)	V(e)
1	-1274.34	28.62580	33.60718
2	-1274.17	23.65982	38.22495
3	-1274.04	24.18239	37.80228
4	-1274.04	24.13439	37.84795
5	-1274.04	24.13923	37.84342
6	-1274.04	24.13875	37.84388

Log-likelihood ratio converged.

Summary result of REML analysis:

Source	Variance	SE
V(G1)	18.203433	7.809399
V(G2)	10.802098	10.414432
V(e)	32.817386	8.614963

Vp	61.822917	4.024798
V(G1)/Vp	0.294445	0.121720
V(G2)/Vp	0.174726	0.167702
Sum of V(G)/Vp	0.469171	0.141402

Sampling variance/covariance of the estimates of variance components:

6.098672e+01	-4.173938e+01	-9.459466e+00
-4.173938e+01	1.084604e+02	-6.253400e+01
-9.459466e+00	-6.253400e+01	7.421758e+01

Summary result of REML analysis has been saved in the file [output/GCTAherit6andnot6.hsq].

Analysis finished at 05:42:23 UTC on Thu Sep 10 2026

Overall computational time: 0.14 sec.

5. GCTA fastGWA: sparse-GRM mixed model for biobank-scale data

fastGWA is GCTA's alternative MLM-based GWAS method, designed for biobank-scale datasets (e.g. UK Biobank) — we apply it here to our much smaller dataset for comparison. The dense GRM used by `--mlma` and GREML above stores every pairwise relationship, so it scales quadratically with sample size — impractical once you're into the hundreds of thousands of individuals. fastGWA's sparse-GRM approximation is what makes mixed-model association testing feasible at that scale; `--fastGWA-mlm` (the command used below) is the actual association test, paired with the sparse GRM built in this section.

Sparse GRM

A sparse GRM keeps only relationship pairs above a cutoff (here 0.05, zeroing out everything below it), which is far cheaper to work with at scale than the full dense GRM — in a large cohort, most pairs of individuals are unrelated (relatedness ≈ 0) anyway, so storing only the meaningfully nonzero entries avoids computing and storing an $N \times N$ matrix that's mostly noise.

```
In [8]: # Keep only relationship pairs > 0.05; everything below that is treated as 0
gcta64 --grm output/GCTAgrm --make-bK-sparse 0.05 --out output/GCTAsparsegrm
```

```
*****
* Genome-wide Complex Trait Analysis (GCTA)
* version v1.95.3 Linux
* Built at Sep  9 2026 10:56:04, by GCC 12.2
* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
*****
Analysis started at 05:42:24 UTC on Thu Sep 10 2026.
Hostname: 8768ca282dde
```

Options:

```
--grm output/GCTAgrm
--make-bK-sparse 0.05
--out output/GCTAsparsegrm
```

```
Pruning the GRM to a sparse matrix with a cutoff of 0.050000...
Total number of parts to be processed: 1
    Saving 498 individual IDs
    Processing part 1
Saving the sparse GRM (894 pairs) to [output/GCTAsparsegrm.grm.sp]
finished generating a sparse GRM Success:
```

```
Analysis finished at 05:42:24 UTC on Thu Sep 10 2026
Overall computational time: 0.00 sec.
```

Partitioned GRM computation

For application using real biobank-scale data, computing the full dense GRM in one pass can be computationally prohibitive. GCTA can partition the computation into m parts (by row) and compute each part separately with `--make-grm-part  $m$   $i$` , to be merged back together afterward. Here we split into 3 parts (using 5 threads each) purely to demonstrate the workflow, then merge the parts and build a second sparse GRM from the result — its top lines should match `GCTAsparsegrm.grm.sp` from above.

```
In [9]: # Compute the GRM in 3 separate row-partitions (parts 1-3 of 3, 5 threads ea
# all three must finish before they're merged in the next cell
gcta64 --bfile ./data/quantfamdata --make-grm-part 3 1 --thread-num 5 \
--out output/test
gcta64 --bfile ./data/quantfamdata --make-grm-part 3 2 --thread-num 5 \
--out output/test
gcta64 --bfile ./data/quantfamdata --make-grm-part 3 3 --thread-num 5 \
--out output/test
```

```
*****
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* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
*****
Analysis started at 05:42:24 UTC on Thu Sep 10 2026.
Hostname: 8768ca282dde
```

Options:

```
--bfile ./data/quantfamdata
--make-grm-part 3 1
--thread-num 5
--out output/test
```

```
The program will be running with up to 5 threads.
Note: GRM is computed using the SNPs on the autosomes.
Reading PLINK FAM file from [./data/quantfamdata.fam]...
498 individuals to be included from FAM file.
498 individuals to be included. 226 males, 272 females, 0 unknown.
Reading PLINK BIM file from [./data/quantfamdata.bim]...
134946 SNPs to be included from BIM file(s).
Computing the genetic relationship matrix (GRM) v2 ...
Subset 1/3, no. subject 1-249
  249 samples, 134946 markers, 31125 GRM elements
IDs for the GRM file have been saved in the file [output/test.part_3_1.grm.i
d]
Computing GRM...
  100% finished in 0.2 sec
134946 SNPs have been processed.
  Used 134946 valid SNPs.
The GRM computation is completed.
Saving GRM...
GRM has been saved in the file [output/test.part_3_1.grm.bin]
Number of SNPs in each pair of individuals has been saved in the file [outpu
t/test.part_3_1.grm.N.bin]
```

```
Analysis finished at 05:42:24 UTC on Thu Sep 10 2026
Overall computational time: 0.30 sec.
```

```
*****
* Genome-wide Complex Trait Analysis (GCTA)
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* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
*****
Analysis started at 05:42:24 UTC on Thu Sep 10 2026.
Hostname: 8768ca282dde
```

Options:

```
--bfile ./data/quantfamdata
--make-grm-part 3 2
--thread-num 5
```

--out output/test

The program will be running with up to 5 threads.

Note: GRM is computed using the SNPs on the autosomes.

Reading PLINK FAM file from [./data/quantfamdata.fam]...

498 individuals to be included from FAM file.

498 individuals to be included. 226 males, 272 females, 0 unknown.

Reading PLINK BIM file from [./data/quantfamdata.bim]...

134946 SNPs to be included from BIM file(s).

Computing the genetic relationship matrix (GRM) v2 ...

Subset 2/3, no. subject 250-390

141 samples, 134946 markers, 45120 GRM elements

IDs for the GRM file have been saved in the file [output/test.part_3_2.grm.id]

Computing GRM...

100% finished in 1.5 sec

134946 SNPs have been processed.

Used 134946 valid SNPs.

The GRM computation is completed.

Saving GRM...

GRM has been saved in the file [output/test.part_3_2.grm.bin]

Number of SNPs in each pair of individuals has been saved in the file [output/test.part_3_2.grm.N.bin]

Analysis finished at 05:42:26 UTC on Thu Sep 10 2026

Overall computational time: 1.58 sec.

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* (C) 2010-present, Yang Lab, Westlake University

* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>

Analysis started at 05:42:26 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Options:

--bfile ./data/quantfamdata

--make-grm-part 3 3

--thread-num 5

--out output/test

The program will be running with up to 5 threads.

Note: GRM is computed using the SNPs on the autosomes.

Reading PLINK FAM file from [./data/quantfamdata.fam]...

498 individuals to be included from FAM file.

498 individuals to be included. 226 males, 272 females, 0 unknown.

Reading PLINK BIM file from [./data/quantfamdata.bim]...

134946 SNPs to be included from BIM file(s).

Computing the genetic relationship matrix (GRM) v2 ...

Subset 3/3, no. subject 391-498

108 samples, 134946 markers, 48006 GRM elements

IDs for the GRM file have been saved in the file [output/test.part_3_3.grm.id]

Computing GRM...

100% finished in 0.6 sec
134946 SNPs have been processed.
Used 134946 valid SNPs.
The GRM computation is completed.
Saving GRM...
GRM has been saved in the file [output/test.part_3_3.grm.bin]
Number of SNPs in each pair of individuals has been saved in the file [output/test.part_3_3.grm.N.bin]

Analysis finished at 05:42:27 UTC on Thu Sep 10 2026
Overall computational time: 0.68 sec.

```
In [10]: # Each partition writes its own output/test.part_3_<i> files; GCTA's GRM for
# just stacked per-pair records, so concatenating the parts reassembles the
cat output/test.part_3_*.grm.id > output/test.grm.id
cat output/test.part_3_*.grm.bin > output/test.grm.bin
cat output/test.part_3_*.grm.N.bin > output/test.grm.N.bin
```

```
In [11]: # Build a sparse GRM from the merged (partitioned) GRM – should match GCTA's
gcta64 --grm output/test --make-bK-sparse 0.05 --out output/newsparsegrm

head output/*.sp # preview both sparse GRMs' top lines to compare
```



```
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* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
*****
Analysis started at 05:42:27 UTC on Thu Sep 10 2026.
Hostname: 8768ca282dde
```

Options:

```
--grm output/test
--make-bK-sparse 0.05
--out output/newsparsegrm
```

Pruning the GRM to a sparse matrix with a cutoff of 0.050000...

Total number of parts to be processed: 1

Saving 498 individual IDs

Processing part 1

Saving the sparse GRM (894 pairs) to [output/newsparsegrm.grm.sp]

finished generating a sparse GRM Success:

Analysis finished at 05:42:27 UTC on Thu Sep 10 2026

Overall computational time: 0.00 sec.

==> output/GCTAsparsegrm.grm.sp <==

0	0	0.98113006
1	0	0.10253318
1	1	0.9935286
2	2	1.0000603
3	2	0.11077
3	3	0.99785215
4	1	0.18634462
4	2	0.11245476
4	3	0.2122499
4	4	0.99371797

==> output/newsparsegrm.grm.sp <==

0	0	0.98112994
1	0	0.10253316
1	1	0.99352795
2	2	1.0000604
3	2	0.11076996
3	3	0.99785256
4	1	0.18634458
4	2	0.11245478
4	3	0.21224986
4	4	0.99371779

Principal components and fastGWA-mlm

For the association analysis, we'd normally use the sparse GRM to model close relationships as random effects, with principal components included as fixed effects.

First compute the top 5 PCs:

```
In [12]: # Compute the top 5 principal components from the (dense) GRM
gcta64 --grm-bin output/GCTAgrm --pca 5 --out output/pcs
```

```
*****
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* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
*****
Analysis started at 05:42:28 UTC on Thu Sep 10 2026.
Hostname: 8768ca282dde
```

```
Accepted options:
--grm-bin output/GCTAgrm
--pca 5
--out output/pcs
```

Note: This is a multi-thread program. You could specify the number of threads by the `--thread-num` option to speed up the computation if there are multiple processors in your machine.

```
Reading IDs of the GRM from [output/GCTAgrm.grm.id].
498 IDs are read from [output/GCTAgrm.grm.id].
Reading the GRM from [output/GCTAgrm.grm.bin].
GRM for 498 individuals are included from [output/GCTAgrm.grm.bin].
```

```
Performing principal component analysis ...
Eigenvalues of 498 individuals have been saved in [output/pcs.eigenval].
The first 5 eigenvectors of 498 individuals have been saved in [output/pcs.eigenvec].
```

```
Analysis finished at 05:42:28 UTC on Thu Sep 10 2026
Overall computational time: 0.16 sec.
```

Now run fastGWA-mlm four ways: with each of the two sparse GRMs (`GCTAsparsegrm` and `newsparsegrm`), both with and without the PCs as covariates. Including the PCs (`--qcovar pcs.eigenvec`) tends to make the genetic variance estimate V_g non-significant, causing fastGWA to silently fall back to plain linear regression — not what we want here — so the no-covariate runs are the ones that actually force a linear mixed model.

```
In [13]: # 1. GCTAsparsegrm + PC covariates
gcta64 --bfile ./data/quantfamdata --grm-sparse output/GCTAsparsegrm \
--fastGWA-mlm --pheno ./data/myphenos.txt --qcovar output/pcs.eigenvec \
--out output/sparse_assoc

# 2. newsparsegrm (partitioned/merged version) + PC covariates
gcta64 --bfile ./data/quantfamdata --grm-sparse output/newsparsegrm \
--fastGWA-mlm --pheno ./data/myphenos.txt --qcovar output/pcs.eigenvec \
--out output/newsparse_assoc

# 3. GCTAsparsegrm, no covariates – forces an actual mixed model (see note a
gcta64 --bfile ./data/quantfamdata --grm-sparse output/GCTAsparsegrm \
```

```
--fastGWA-mlm --pheno ./data/myphenos.txt --out output/LMMsparse_assoc
```

```
# 4. newsparsegrm, no covariates
```

```
gcta64 --bfile ./data/quantfamdata --grm-sparse output/newsparsegrm \
```

```
--fastGWA-mlm --pheno ./data/myphenos.txt --out output/LMMnewsparse_assoc
```

```
*****
* Genome-wide Complex Trait Analysis (GCTA)
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* (C) 2010-present, Yang Lab, Westlake University
* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>
*****
Analysis started at 05:42:28 UTC on Thu Sep 10 2026.
Hostname: 8768ca282dde
```

Options:

```
--bfile ./data/quantfamdata
--grm-sparse output/GCTAsparsegrm
--fastGWA-mlm
--pheno ./data/myphenos.txt
--qcovar output/pcs.eigenvec
--out output/sparse_assoc
```

```
Reading PLINK FAM file from [./data/quantfamdata.fam]...
498 individuals to be included from FAM file.
Reading phenotype data from [./data/myphenos.txt]...
498 overlapping individuals with non-missing data to be included from the ph
enotype file.
498 individuals to be included. 226 males, 272 females, 0 unknown.
Reading PLINK BIM file from [./data/quantfamdata.bim]...
134946 SNPs to be included from BIM file(s).
Reading quantitative covariates from [output/pcs.eigenvec].
5 covariates of 498 samples to be included.
498 overlapping individuals with non-missing data to be included from the co
variate file(s).
Reading the sparse GRM file from [output/GCTAsparsegrm]...
After matching all the files, 498 individuals to be included in the analysi
s.
Estimating the genetic variance (Vg) by fastGWA-REML (grid search)...
Iteration 1, step size: 0.974355, logL: -1270.8. Vg: 20.4615, searching rang
e: 19.4871 to 21.4358
Iteration 2, step size: 0.129914, logL: -1270.8. Vg: 20.3965, searching rang
e: 20.2666 to 20.5264
Iteration 3, step size: 0.0173219, logL: -1270.8. Vg: 20.3532, searching ran
ge: 20.3359 to 20.3705
Iteration 4, step size: 0.00230958, logL: -1270.8. Vg: 20.359, searching ran
ge: 20.3567 to 20.3613
Iteration 5, step size: 0.000307944, logL: -1270.8. Vg: 20.3597, searching r
ange: 20.3594 to 20.36
Iteration 6, step size: 4.10592e-05, logL: -1270.8. Vg: 20.3597, searching r
ange: 20.3596 to 20.3597
Iteration 7, step size: 5.47457e-06, logL: -1270.8. Vg: 20.3597, searching r
ange: 20.3597 to 20.3597
Iteration 8, step size: 7.29942e-07, logL: -1270.8. Vg: 20.3597, searching r
ange: 20.3597 to 20.3597
Iteration 9, step size: 9.73256e-08, logL: -1270.8. Vg: 20.3597, searching r
ange: 20.3597 to 20.3597
Iteration 10, step size: 1.29767e-08, logL: -1270.8. Vg: 20.3597, searching
range: 20.3597 to 20.3597
Iteration 11, step size: 1.73023e-09, logL: -1270.8. Vg: 20.3597, searching
```

range: 20.3597 to 20.3597
 Iteration 12, step size: 2.30698e-10, logL: -1270.8. Vg: 20.3597, searching
 range: 20.3597 to 20.3597
 Iteration 13, step size: 3.07596e-11, logL: -1270.8. Vg: 20.3597, searching
 range: 20.3597 to 20.3597
 fastGWA-REML converged.
 logL: -1270.8
 Sampling variance/covariance of the estimates of Vg and Ve:
 113.789 -103.598
 -103.598 108.573

Source	Variance	SE
Vg	20.3597	10.6672
Ve	40.5375	10.4198
Vp	60.8972	

Heritability = 0.334328 (Pval = 0.0563109)
 fastGWA-REML runtime: 0.0317692 sec.

Warning: the estimate of Vg is not statistically significant (i.e., $p > 0.05$). This is likely because the number of closely related individuals in the sample is not large enough.

In this case, the program will use linear regression for association test.

Performing fastGWA linear regression analysis...

fastGWA results will be saved in text format to [output/sparse_assoc.fastGWA].

Filtering out variants with MAF < 0.0001, or customise it with --maf flag.

Filtering out variants with missingness rate > 0.10, or customise it with --geno flag.

100% finished in 0.7 sec

134946 SNPs have been processed.

Saved 134946 SNPs.

Analysis finished at 05:42:29 UTC on Thu Sep 10 2026

Overall computational time: 0.82 sec.

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* version v1.95.3 Linux

* Built at Sep 9 2026 10:56:04, by GCC 12.2

* (C) 2010-present, Yang Lab, Westlake University

* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>

Analysis started at 05:42:29 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Options:

--bfile ./data/quantfamdata

--grm-sparse output/newsparsegrm

--fastGWA-mlm

--pheno ./data/myphenos.txt

--qcovar output/pes.eigenvec

--out output/newsparse_assoc

Reading PLINK FAM file from [./data/quantfamdata.fam]...

498 individuals to be included from FAM file.

Reading phenotype data from [./data/myphenos.txt]...
 498 overlapping individuals with non-missing data to be included from the phenotype file.
 498 individuals to be included. 226 males, 272 females, 0 unknown.
 Reading PLINK BIM file from [./data/quantfamdata.bim]...
 134946 SNPs to be included from BIM file(s).
 Reading quantitative covariates from [output/pcs.eigenvec].
 5 covariates of 498 samples to be included.
 498 overlapping individuals with non-missing data to be included from the covariate file(s).
 Reading the sparse GRM file from [output/newsparsegrm]...
 After matching all the files, 498 individuals to be included in the analysis.
 Estimating the genetic variance (V_g) by fastGWA-REML (grid search)...
 Iteration 1, step size: 0.974355, logL: -1270.8. V_g : 20.4615, searching range: 19.4871 to 21.4358
 Iteration 2, step size: 0.129914, logL: -1270.8. V_g : 20.3965, searching range: 20.2666 to 20.5264
 Iteration 3, step size: 0.0173219, logL: -1270.8. V_g : 20.3532, searching range: 20.3359 to 20.3705
 Iteration 4, step size: 0.00230958, logL: -1270.8. V_g : 20.359, searching range: 20.3567 to 20.3613
 Iteration 5, step size: 0.000307944, logL: -1270.8. V_g : 20.3597, searching range: 20.3594 to 20.36
 Iteration 6, step size: 4.10592e-05, logL: -1270.8. V_g : 20.3597, searching range: 20.3596 to 20.3597
 Iteration 7, step size: 5.47457e-06, logL: -1270.8. V_g : 20.3597, searching range: 20.3597 to 20.3597
 Iteration 8, step size: 7.29942e-07, logL: -1270.8. V_g : 20.3597, searching range: 20.3597 to 20.3597
 Iteration 9, step size: 9.73256e-08, logL: -1270.8. V_g : 20.3597, searching range: 20.3597 to 20.3597
 Iteration 10, step size: 1.29767e-08, logL: -1270.8. V_g : 20.3597, searching range: 20.3597 to 20.3597
 Iteration 11, step size: 1.73023e-09, logL: -1270.8. V_g : 20.3597, searching range: 20.3597 to 20.3597
 Iteration 12, step size: 2.30698e-10, logL: -1270.8. V_g : 20.3597, searching range: 20.3597 to 20.3597
 Iteration 13, step size: 3.07596e-11, logL: -1270.8. V_g : 20.3597, searching range: 20.3597 to 20.3597
 fastGWA-REML converged.
 logL: -1270.8
 Sampling variance/covariance of the estimates of V_g and V_e :
 113.789 -103.598
 -103.598 108.573

Source	Variance	SE
V_g	20.3597	10.6672
V_e	40.5375	10.4198
V_p	60.8972	

Heritability = 0.334329 (Pval = 0.0563108)
 fastGWA-REML runtime: 0.0457185 sec.

Warning: the estimate of V_g is not statistically significant (i.e., $p > 0.05$). This is likely because the number of closely related individuals in the sample is not large enough.

In this case, the program will use linear regression for association test.

Performing fastGWA linear regression analysis...

fastGWA results will be saved in text format to [output/newsparse_assoc.fastGWA].

Filtering out variants with MAF < 0.0001, or customise it with --maf flag.

Filtering out variants with missingness rate > 0.10, or customise it with --geno flag.

100% finished in 0.7 sec

134946 SNPs have been processed.

Saved 134946 SNPs.

Analysis finished at 05:42:30 UTC on Thu Sep 10 2026

Overall computational time: 0.84 sec.

* Genome-wide Complex Trait Analysis (GCTA)

* version v1.95.3 Linux

* Built at Sep 9 2026 10:56:04, by GCC 12.2

* (C) 2010-present, Yang Lab, Westlake University

* Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>

Analysis started at 05:42:30 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Options:

--bfile ./data/quantfamdata

--grm-sparse output/GCTAsparsegrm

--fastGWA-mlm

--pheno ./data/myphenos.txt

--out output/LMMsparse_assoc

Reading PLINK FAM file from [./data/quantfamdata.fam]...

498 individuals to be included from FAM file.

Reading phenotype data from [./data/myphenos.txt]...

498 overlapping individuals with non-missing data to be included from the phenotype file.

498 individuals to be included. 226 males, 272 females, 0 unknown.

Reading PLINK BIM file from [./data/quantfamdata.bim]...

134946 SNPs to be included from BIM file(s).

Reading the sparse GRM file from [output/GCTAsparsegrm]...

After matching all the files, 498 individuals to be included in the analysis.

Estimating the genetic variance (Vg) by fastGWA-REML (grid search)...

Iteration 1, step size: 0.996879, logL: -1273.59. Vg: 27.9126, searching range: 26.9157 to 28.9095

Iteration 2, step size: 0.132917, logL: -1273.59. Vg: 28.3778, searching range: 28.2449 to 28.5108

Iteration 3, step size: 0.0177223, logL: -1273.59. Vg: 28.3158, searching range: 28.2981 to 28.3335

Iteration 4, step size: 0.00236297, logL: -1273.59. Vg: 28.3217, searching range: 28.3194 to 28.3241

Iteration 5, step size: 0.000315063, logL: -1273.59. Vg: 28.3216, searching range: 28.3212 to 28.3219

Iteration 6, step size: 4.20084e-05, logL: -1273.59. Vg: 28.3215, searching range: 28.3215 to 28.3215

Iteration 7, step size: 5.60112e-06, logL: -1273.59. Vg: 28.3215, searching range: 28.3215 to 28.3215
 Iteration 8, step size: 7.46816e-07, logL: -1273.59. Vg: 28.3215, searching range: 28.3215 to 28.3215
 Iteration 9, step size: 9.95755e-08, logL: -1273.59. Vg: 28.3215, searching range: 28.3215 to 28.3215
 Iteration 10, step size: 1.32767e-08, logL: -1273.59. Vg: 28.3215, searching range: 28.3215 to 28.3215
 Iteration 11, step size: 1.77023e-09, logL: -1273.59. Vg: 28.3215, searching range: 28.3215 to 28.3215
 Iteration 12, step size: 1.18015e-10, logL: -1273.59. Vg: 28.3215, searching range: 28.3215 to 28.3215
 Iteration 13, step size: 7.8676e-12, logL: -1273.59. Vg: 28.3215, searching range: 28.3215 to 28.3215
 fastGWA-REML converged.
 logL: -1273.59
 Sampling variance/covariance of the estimates of Vg and Ve:
 108.552 -94.7786
 -94.7786 97.1234

Source	Variance	SE
Vg	28.3215	10.4188
Ve	33.9835	9.85512
Vp	62.305	

Heritability = 0.454562 (Pval = 0.00656192)
 fastGWA-REML runtime: 0.0190891 sec.

Tuning parameters using 2000 null SNPs...
 reading genotypes...
 100% finished in 0.1 sec
 1981 SNPs have been processed.
 Tuning of Gamma finished at the 1000th SNP.
 Mean GRAMMAR-Gamma value = 0.015448
 Tuning of Gamma finished 0.089907 seconds.

Performing fastGWA mixed model association analysis...
 fastGWA results will be saved in text format to [output/LMMsparse_assoc.fastGWA].

Filtering out variants with MAF < 0.0001, or customise it with --maf flag.
 Filtering out variants with missingness rate > 0.10, or customise it with --geno flag.
 100% finished in 0.6 sec
 134946 SNPs have been processed.
 Saved 134946 SNPs.

Analysis finished at 05:42:31 UTC on Thu Sep 10 2026
 Overall computational time: 0.85 sec.

 * Genome-wide Complex Trait Analysis (GCTA)
 * version v1.95.3 Linux
 * Built at Sep 9 2026 10:56:04, by GCC 12.2
 * (C) 2010-present, Yang Lab, Westlake University
 * Please report bugs to Jian Yang <jian.yang@westlake.edu.cn>

 Analysis started at 05:42:31 UTC on Thu Sep 10 2026.

Hostname: 8768ca282dde

Options:

```
--bfile ./data/quantfamdata
--grm-sparse output/newsparsegrm
--fastGWA-mlm
--pheno ./data/myphenos.txt
--out output/LMMnewsparse_assoc
```

```
Reading PLINK FAM file from [./data/quantfamdata.fam]...
498 individuals to be included from FAM file.
Reading phenotype data from [./data/myphenos.txt]...
498 overlapping individuals with non-missing data to be included from the ph
enotype file.
498 individuals to be included. 226 males, 272 females, 0 unknown.
Reading PLINK BIM file from [./data/quantfamdata.bim]...
134946 SNPs to be included from BIM file(s).
Reading the sparse GRM file from [output/newsparsegrm]...
After matching all the files, 498 individuals to be included in the analysi
s.
Estimating the genetic variance (Vg) by fastGWA-REML (grid search)...
Iteration 1, step size: 0.996879, logL: -1273.59. Vg: 27.9126, searching ran
ge: 26.9157 to 28.9095
Iteration 2, step size: 0.132917, logL: -1273.59. Vg: 28.3778, searching ran
ge: 28.2449 to 28.5108
Iteration 3, step size: 0.0177223, logL: -1273.59. Vg: 28.3158, searching ra
nge: 28.2981 to 28.3335
Iteration 4, step size: 0.00236297, logL: -1273.59. Vg: 28.3217, searching r
ange: 28.3194 to 28.3241
Iteration 5, step size: 0.000315063, logL: -1273.59. Vg: 28.3216, searching
range: 28.3212 to 28.3219
Iteration 6, step size: 4.20084e-05, logL: -1273.59. Vg: 28.3215, searching
range: 28.3215 to 28.3215
Iteration 7, step size: 5.60112e-06, logL: -1273.59. Vg: 28.3215, searching
range: 28.3215 to 28.3215
Iteration 8, step size: 7.46816e-07, logL: -1273.59. Vg: 28.3215, searching
range: 28.3215 to 28.3215
Iteration 9, step size: 9.95755e-08, logL: -1273.59. Vg: 28.3215, searching
range: 28.3215 to 28.3215
Iteration 10, step size: 6.63837e-09, logL: -1273.59. Vg: 28.3215, searching
range: 28.3215 to 28.3215
Iteration 11, step size: 8.85116e-10, logL: -1273.59. Vg: 28.3215, searching
range: 28.3215 to 28.3215
Iteration 12, step size: 1.18015e-10, logL: -1273.59. Vg: 28.3215, searching
range: 28.3215 to 28.3215
Iteration 13, step size: 1.57352e-11, logL: -1273.59. Vg: 28.3215, searching
range: 28.3215 to 28.3215
fastGWA-REML converged.
logL: -1273.59
Sampling variance/covariance of the estimates of Vg and Ve:
 108.552 -94.7786
-94.7786  97.1234
```

Source	Variance	SE
Vg	28.3215	10.4188

```
Ve      33.9835 9.85512
Vp      62.305
```

```
Heritability = 0.454562 (Pval = 0.0065619)
fastGWA-REML runtime: 0.0184981 sec.
```

```
Tuning parameters using 2000 null SNPs...
  reading genotypes...
  100% finished in 0.1 sec
1981 SNPs have been processed.
Tuning of Gamma finished at the 1000th SNP.
Mean GRAMMAR-Gamma value = 0.015448
Tuning of Gamma finished 0.0906088 seconds.
```

```
Performing fastGWA mixed model association analysis...
fastGWA results will be saved in text format to [output/LMMnewsparse_assoc.f
astGWA].
  Filtering out variants with MAF < 0.0001, or customise it with --maf flag.
  Filtering out variants with missingness rate > 0.10, or customise it with
--geno flag.
  100% finished in 0.7 sec
134946 SNPs have been processed.
Saved 134946 SNPs.
```

```
Analysis finished at 05:42:32 UTC on Thu Sep 10 2026
Overall computational time: 0.90 sec.
```

Part 2 — Reading Results and Plotting

Switch kernels now: in JupyterLab's menu bar, go to **Kernel → Change Kernel... → R**, then continue running cells from here. Everything below reads files already written by Part 1, so nothing needs to be re-run in Bash.

Kernel: run the code cells in this section with the **R** kernel.

```
In [1]: # Load the manhattan()/qq() plotting helper functions used throughout Part 2
source("scripts/qqmanHJCupdated.R")
```

PLINK results

134,946 rows, one per SNP — too many to eyeball, so we visualize with a Manhattan plot (colored by chromosome) and a QQ plot. Visually, chromosomes 6 and 12 stand out, with possibly chromosome 5 too.

The genomic control inflation factor λ converts p-values to 1-df chi-squared statistics and takes their median, scaled by 0.456 (Devlin & Roeder, 1999) — values near 1.0

indicate the bulk of results are consistent with the null. Expect $\lambda \approx 1.10$ here (mild inflation from relatedness, since plain PLINK doesn't account for it).

```
In [2]: # Read PLINK's association results and preview the columns
res1 <- read.table("output/plinkresults.qassoc", header = T)
head(res1)
```

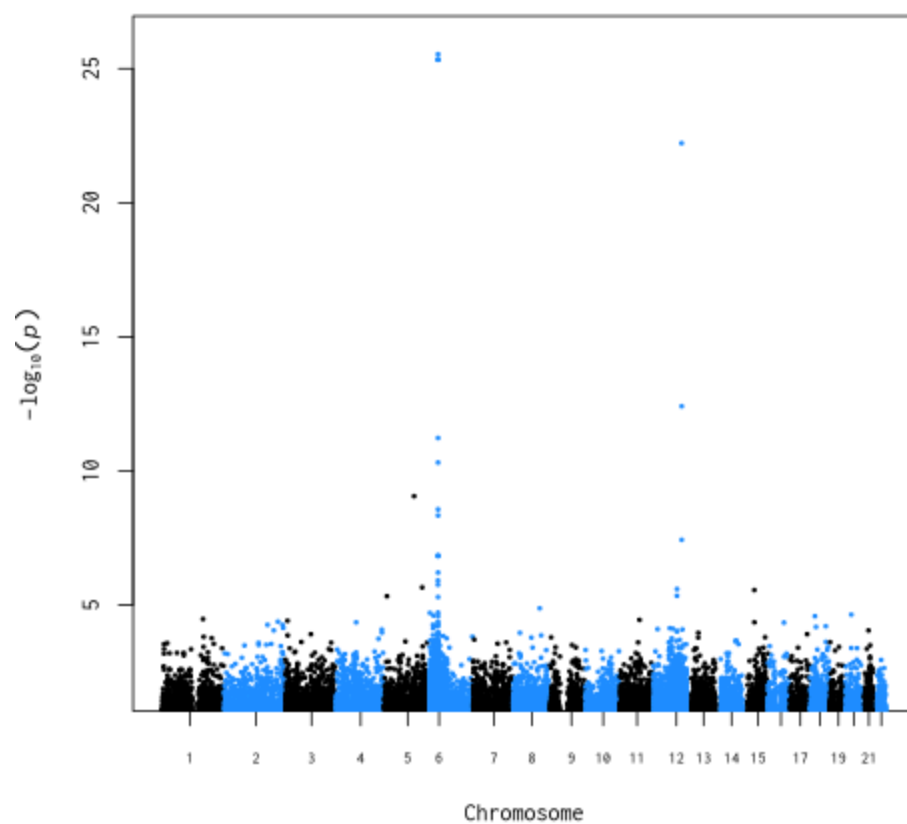
A data.frame: 6 × 9

	CHR	SNP	BP	NMISS	BETA	SE	R2	T	P
	<int>	<chr>	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1	rs9326624	558615	498	0.526900	0.9159	6.668e-04	0.57530	0.5654
2	1	rs3094315	742429	498	0.306700	0.5377	6.555e-04	0.57040	0.5687
3	1	rs12562034	758311	495	0.793000	0.8845	1.628e-03	0.89650	0.3704
4	1	rs3934834	995669	498	-0.007573	0.6342	2.875e-07	-0.01194	0.9905
5	1	rs3737728	1011278	498	0.465700	0.6392	1.069e-03	0.72860	0.4666
6	1	rs4970405	1038818	498	-0.829400	0.7574	2.412e-03	-1.09500	0.2740

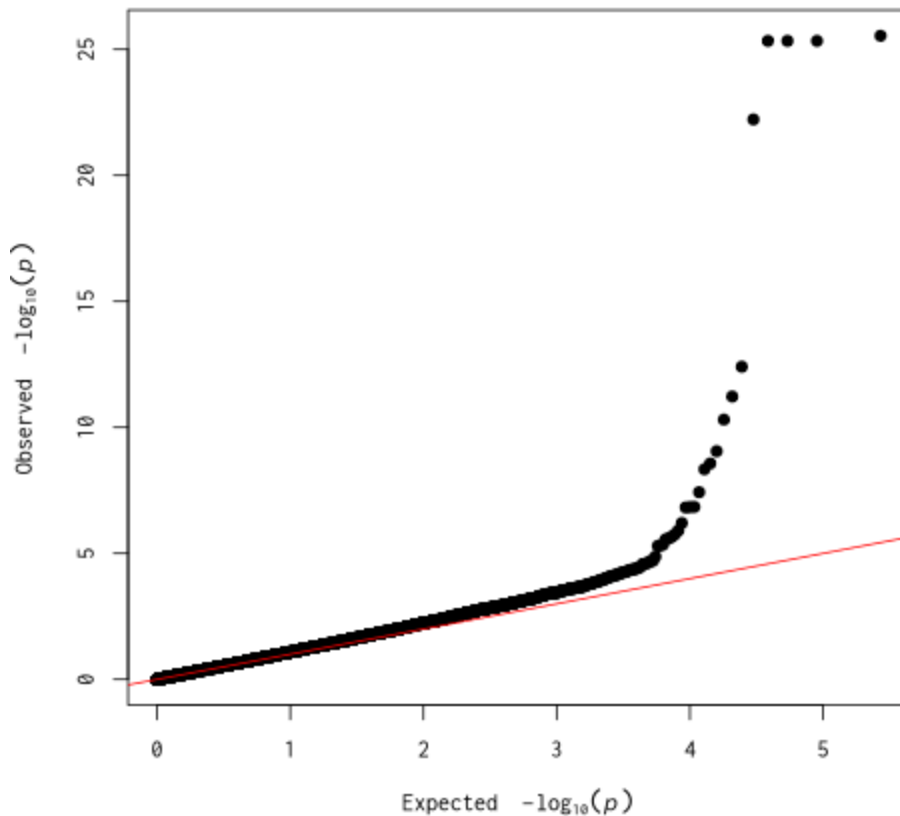
```
In [3]: # Manhattan plot: -log10(P) by genomic position, colored by chromosome (optional)
png("output/mh1.png")
manhattan(res1,
  pch = 20, suggestiveline = F, genomewideline = F, ymin = 2,
  cex.x.axis = 0.65, colors = c("black", "dodgerblue"), cex = 0.5
)
dev.off()
IRdisplay::display_png(file = "output/mh1.png")

# QQ plot: observed vs. expected -log10(P) under the null (should hug the diagonal)
png("output/qq1.png")
qq(res1$P)
dev.off()
IRdisplay::display_png(file = "output/qq1.png")
```

pdf: 2



pdf: 2



```
In [4]: # Genomic control inflation factor: convert p-values to 1-df chi-squared sta
# take the median, and scale by the expected null median (0.456) – see note
chi <- (qchisq(1 - res1$P, 1))
lambda <- median(chi) / 0.456
lambda
```

1.1024932733801

FaST-LMM results

The p-value column here is named `Pvalue`, and results are already ordered by significance. Expect $\lambda \approx 0.99$ — noticeably less inflated than PLINK's regression, since FaST-LMM accounts for relatedness. (This run doesn't use leave-one-chromosome-out, matching the original exercise's own `fastlmmc` invocation — see the note in Part 1.)

```
In [5]: # Read FaST-LMM's association results (tab-delimited) and preview the column
res2 <- read.table("output/FLMMresults", header = T, sep = "\t")
head(res2)
```

	SNP	Chromosome	GeneticDistance	Position	Pvalue	Qvalue	
	<chr>	<int>	<dbl>	<int>	<dbl>	<dbl>	<int>
1	rs28366284	6	0	32668305	6.272724e-25	2.489451e-20	49
2	rs9271209	6	0	32687013	7.379104e-25	2.489451e-20	49
3	rs9271252	6	0	32688266	7.379104e-25	2.489451e-20	49
4	rs9271255	6	0	32688335	7.379104e-25	2.489451e-20	49
5	rs233722	12	0	111515857	3.921467e-22	1.058373e-17	49
6	rs233716	12	0	111524326	1.666483e-12	3.748087e-08	49

```
In [6]: # Genomic control lambda for FaST-LMM (see the PLINK results section above)
chi <- (qchisq(1 - res2$Pvalue, 1))
lambda <- median(chi) / 0.456
lambda
```

0.991378659519806

```
In [7]: # manhattan()/qq() expect columns named SNP, CHR, BP, P – rename FaST-LMM's
new <- data.frame(res2$SNP, res2$Chromosome, res2$Position, res2$Pvalue)
names(new) <- c("SNP", "CHR", "BP", "P")
head(new)

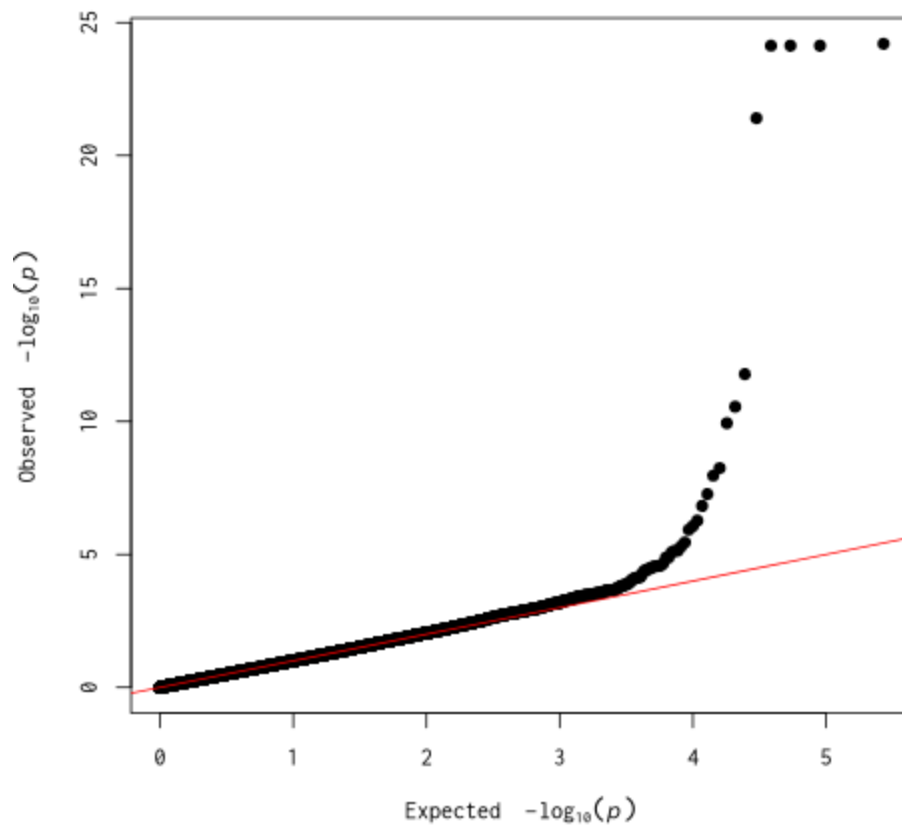
png("output/qq2.png")
qq(new$P)
dev.off()
IRdisplay::display_png(file = "output/qq2.png")

png("output/mh2.png")
manhattan(new,
  pch = 20, suggestiveline = F, genomewideline = F, ymin = 2,
  cex.x.axis = 0.65, colors = c("black", "dodgerblue"), cex = 0.5
)
dev.off()
IRdisplay::display_png(file = "output/mh2.png")
```

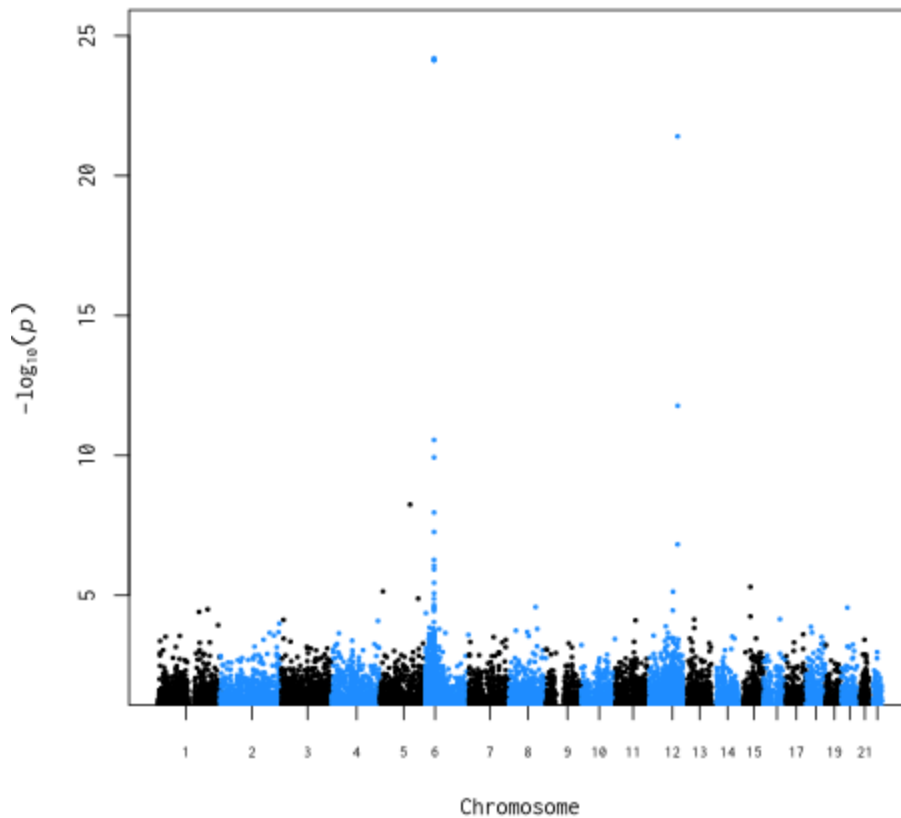
A data.frame: 6 × 4

	SNP	CHR	BP	P
	<chr>	<int>	<int>	<dbl>
1	rs28366284	6	32668305	6.272724e-25
2	rs9271209	6	32687013	7.379104e-25
3	rs9271252	6	32688266	7.379104e-25
4	rs9271255	6	32688335	7.379104e-25
5	rs233722	12	111515857	3.921467e-22
6	rs233716	12	111524326	1.666483e-12

pdf: 2



pdf: 2



GCTA `--mlma` results

Expect λ close to 1.0, and QQ/Manhattan plots similar to FaST-LMM's.

```
In [8]: # Read GCTA --mlma's association results and preview the columns
res3 <- read.table("output/GCTAresults.mlma", header = T)
head(res3)
```

A data.frame: 6 × 9

	Chr	SNP	bp	A1	A2	Freq	b	se	
	<int>	<chr>	<int>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	1	rs9326624	558615	G	A	0.0913655	0.693811	0.924665	0.45305
2	1	rs3094315	742429	G	A	0.3162650	0.494021	0.550792	0.36975
3	1	rs12562034	758311	A	G	0.0818182	0.827582	0.882891	0.34857
4	1	rs3934834	995669	A	G	0.1817270	-0.121518	0.640661	0.84956
5	1	rs3737728	1011278	A	G	0.1957830	0.513895	0.644486	0.42523
6	1	rs4970405	1038818	G	A	0.1194780	-0.697621	0.764350	0.36140


```
In [9]: # Genomic control lambda for GCTA --mlma
chi <- (qchisq(1 - res3$p, 1))
lambda <- median(chi) / 0.456
lambda
```

0.986985057933128

```
In [10]: # Rename to the SNP/CHR/BP/P columns manhattan()/qq() expect (see the FaST-L
new3 <- data.frame(res3$SNP, res3$Chr, res3$bp, res3$p)
names(new3) <- c("SNP", "CHR", "BP", "P")
head(new3)

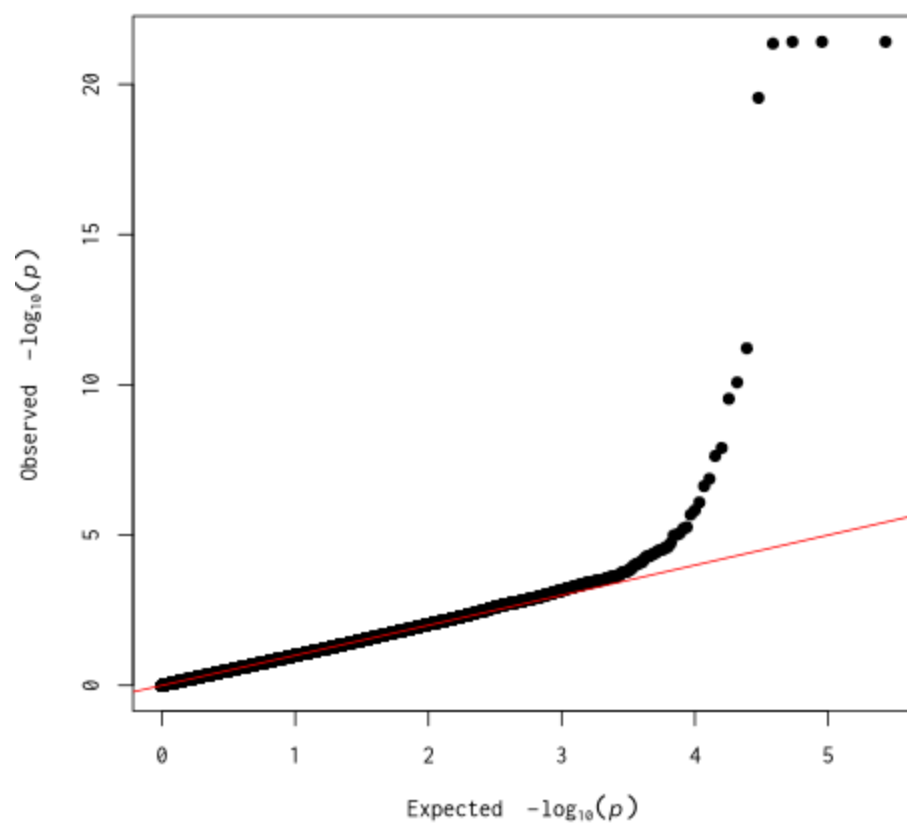
png("output/qq3.png")
qq(new3$P)
dev.off()
IRdisplay::display_png(file = "output/qq3.png")

png("output/mh3.png")
manhattan(new3,
  pch = 20, suggestiveline = F, genomewideline = F, ymin = 2,
  cex.x.axis = 0.65, colors = c("black", "dodgerblue"), cex = 0.5
)
dev.off()
IRdisplay::display_png(file = "output/mh3.png")
```

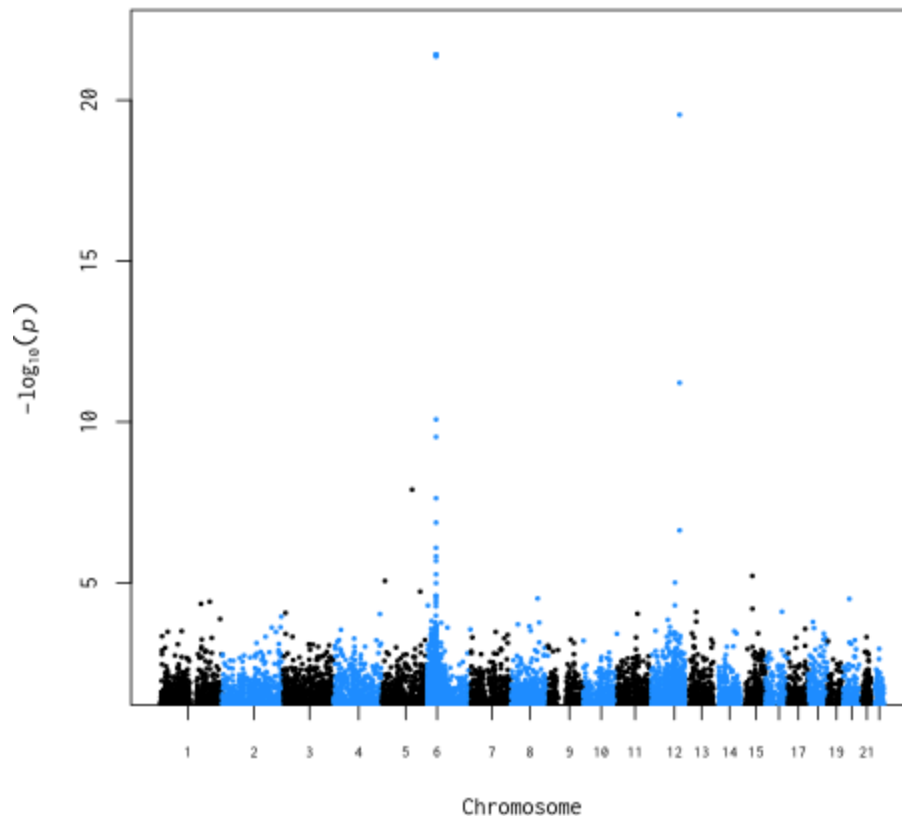
A data.frame: 6 × 4

	SNP	CHR	BP	P
	<chr>	<int>	<int>	<dbl>
1	rs9326624	1	558615	0.453052
2	rs3094315	1	742429	0.369758
3	rs12562034	1	758311	0.348576
4	rs3934834	1	995669	0.849563
5	rs3737728	1	1011278	0.425236
6	rs4970405	1	1038818	0.361401

pdf: 2



pdf: 2



Comparing GCTA to FaST-LMM

Expect the two methods' $-\log_{10}(P)$ values to line up closely along the diagonal — both are mixed models over the same data without a LOCO guard, so any gap here reflects implementation differences (e.g. GCTA's exact vs. FaST-LMM's spectral variance-component estimation) rather than a methodological one.

```
In [11]: # Join GCTA's and FaST-LMM's results on SNP so we can compare p-values for t
new2 <- data.frame(res2$SNP, res2$Chromosome, res2$Position, res2$Pvalue)
names(new2) <- c("SNP", "CHR", "BP", "P")
merged <- merge(new3, new2, by = "SNP", sort = F)

head(res2)
head(new2)
head(new3)
head(merged)
```

	SNP	Chromosome	GeneticDistance	Position	Pvalue	Qvalue	
	<chr>	<int>	<dbl>	<int>	<dbl>	<dbl>	<int>
1	rs28366284	6	0	32668305	6.272724e-25	2.489451e-20	49
2	rs9271209	6	0	32687013	7.379104e-25	2.489451e-20	49
3	rs9271252	6	0	32688266	7.379104e-25	2.489451e-20	49
4	rs9271255	6	0	32688335	7.379104e-25	2.489451e-20	49
5	rs233722	12	0	111515857	3.921467e-22	1.058373e-17	49
6	rs233716	12	0	111524326	1.666483e-12	3.748087e-08	49

A data.frame: 6 x 4

	SNP	CHR	BP	P
	<chr>	<int>	<int>	<dbl>
1	rs28366284	6	32668305	6.272724e-25
2	rs9271209	6	32687013	7.379104e-25
3	rs9271252	6	32688266	7.379104e-25
4	rs9271255	6	32688335	7.379104e-25
5	rs233722	12	111515857	3.921467e-22
6	rs233716	12	111524326	1.666483e-12

A data.frame: 6 x 4

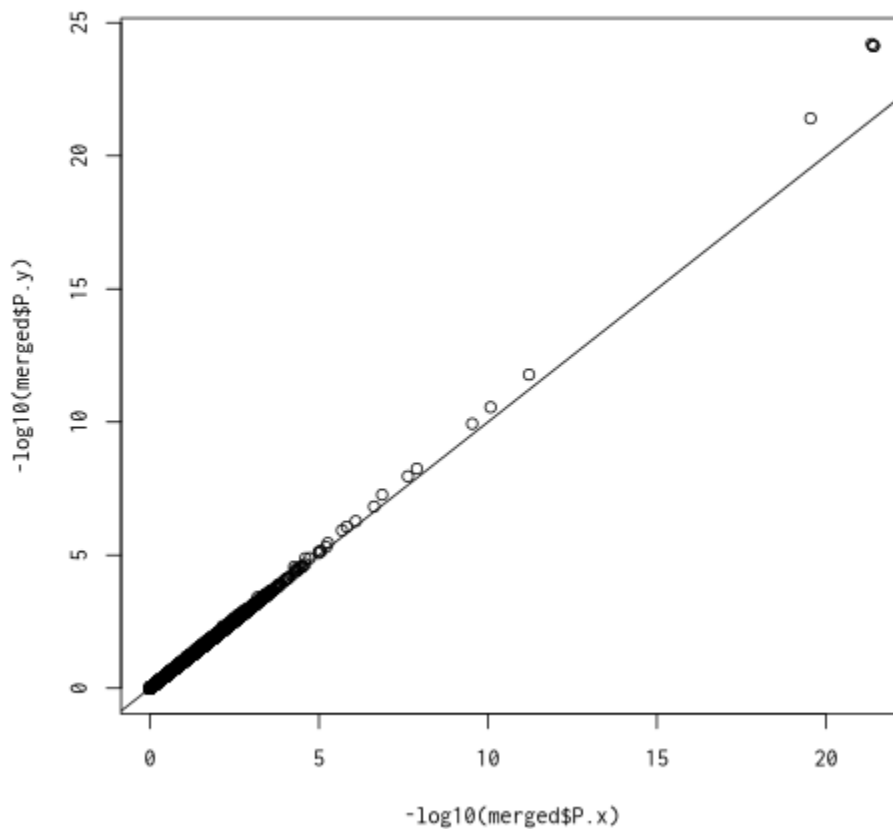
	SNP	CHR	BP	P
	<chr>	<int>	<int>	<dbl>
1	rs9326624	1	558615	0.453052
2	rs3094315	1	742429	0.369758
3	rs12562034	1	758311	0.348576
4	rs3934834	1	995669	0.849563
5	rs3737728	1	1011278	0.425236
6	rs4970405	1	1038818	0.361401

A data.frame: 6 × 7

	SNP	CHR.x	BP.x	P.x	CHR.y	BP.y	P.y
	<chr>	<int>	<int>	<dbl>	<int>	<int>	<dbl>
1	rs9326624	1	558615	0.453052	1	558615	0.4529177
2	rs3094315	1	742429	0.369758	1	742429	0.3683530
3	rs12562034	1	758311	0.348576	1	758311	0.3476341
4	rs3934834	1	995669	0.849563	1	995669	0.8495270
5	rs3737728	1	1011278	0.425236	1	1011278	0.4276619
6	rs4970405	1	1038818	0.361401	1	1038818	0.3597434

```
In [12]: # Scatter of -log10(P): the diagonal line marks perfect agreement (y = x), r
png("output/compareGCTAFLMM.png")
plot(-log10(merged$P.x), -log10(merged$P.y))
abline(0, 1)
dev.off()
IRdisplay::display_png(file = "output/compareGCTAFLMM.png")
```

pdf: 2



GCTA fastGWA results

Four result sets from Part 1: sparse GRM with/without PC covariates, for each of the two (equivalent) sparse GRMs we built. The linear-mixed-model runs (no PC covariates, forced to actually use the sparse GRM) should show λ noticeably closer to 1.0 than the PC-covariate runs, which fastGWA silently downgraded to plain linear regression.

```
In [13]: # res4/res5: sparse GRM + PC covariates (fastGWA silently falls back to line
# res6/res7: same two GRMs, no covariates (forces an actual mixed model)
res4 <- read.table("output/sparse_assoc.fastGWA", header = T)
res5 <- read.table("output/newsparse_assoc.fastGWA", header = T)
res6 <- read.table("output/LMMsparse_assoc.fastGWA", header = T)
res7 <- read.table("output/LMMnewsparse_assoc.fastGWA", header = T)

head(res4)
head(res5)
head(res6)
head(res7)
```

A data.frame: 6 × 10

	CHR	SNP	POS	A1	A2	N	AF1	BETA	SE
	<int>	<chr>	<int>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>
1	1	rs9326624	558615	G	A	498	0.0913655	0.590324	0.905413
2	1	rs3094315	742429	G	A	498	0.3162650	0.490011	0.531356
3	1	rs12562034	758311	A	G	495	0.0818182	0.722313	0.874147
4	1	rs3934834	995669	A	G	498	0.1817270	0.168089	0.626957
5	1	rs3737728	1011278	A	G	498	0.1957830	0.376003	0.632034
6	1	rs4970405	1038818	G	A	498	0.1194780	-0.737276	0.748977

A data.frame: 6 × 10

	CHR	SNP	POS	A1	A2	N	AF1	BETA	SE
	<int>	<chr>	<int>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>
1	1	rs9326624	558615	G	A	498	0.0913655	0.590324	0.905413
2	1	rs3094315	742429	G	A	498	0.3162650	0.490011	0.531356
3	1	rs12562034	758311	A	G	495	0.0818182	0.722313	0.874147
4	1	rs3934834	995669	A	G	498	0.1817270	0.168089	0.626957
5	1	rs3737728	1011278	A	G	498	0.1957830	0.376003	0.632034
6	1	rs4970405	1038818	G	A	498	0.1194780	-0.737276	0.748977

A data.frame: 6 × 10

	CHR	SNP	POS	A1	A2	N	AF1	BETA	SE
	<int>	<chr>	<int>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>
1	1	rs9326624	558615	G	A	498	0.0913655	0.6800840	0.932956
2	1	rs3094315	742429	G	A	498	0.3162650	0.5165540	0.547754
3	1	rs12562034	758311	A	G	495	0.0818182	0.8136900	0.900972
4	1	rs3934834	995669	A	G	498	0.1817270	-0.0701891	0.645799
5	1	rs3737728	1011278	A	G	498	0.1957830	0.4090710	0.651214
6	1	rs4970405	1038818	G	A	498	0.1194780	-0.7021490	0.772184

A data.frame: 6 × 10

	CHR	SNP	POS	A1	A2	N	AF1	BETA	SE
	<int>	<chr>	<int>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>
1	1	rs9326624	558615	G	A	498	0.0913655	0.6800840	0.932956
2	1	rs3094315	742429	G	A	498	0.3162650	0.5165540	0.547754
3	1	rs12562034	758311	A	G	495	0.0818182	0.8136900	0.900972
4	1	rs3934834	995669	A	G	498	0.1817270	-0.0701892	0.645799
5	1	rs3737728	1011278	A	G	498	0.1957830	0.4090710	0.651214
6	1	rs4970405	1038818	G	A	498	0.1194780	-0.7021490	0.772184

```
In [14]: # res4: sparse GRM + PC covariates
chi <- (qchisq(1 - res4$P, 1))
lambda <- median(chi) / 0.456
lambda
# res5: newparsegrm + PC covariates
chi <- (qchisq(1 - res5$P, 1))
lambda <- median(chi) / 0.456
lambda
# res6: sparse GRM, no covariates (forced mixed model)
chi <- (qchisq(1 - res6$P, 1))
lambda <- median(chi) / 0.456
lambda
# res7: newparsegrm, no covariates (forced mixed model)
chi <- (qchisq(1 - res7$P, 1))
lambda <- median(chi) / 0.456
lambda
```

```
1.06102625393192
1.06102625393192
1.00438438440012
1.00438438440012
```

```
In [15]: # res4: sparse GRM + PC covariates
new4 <- data.frame(res4$SNP, res4$CHR, res4$POS, res4$P)
```

```

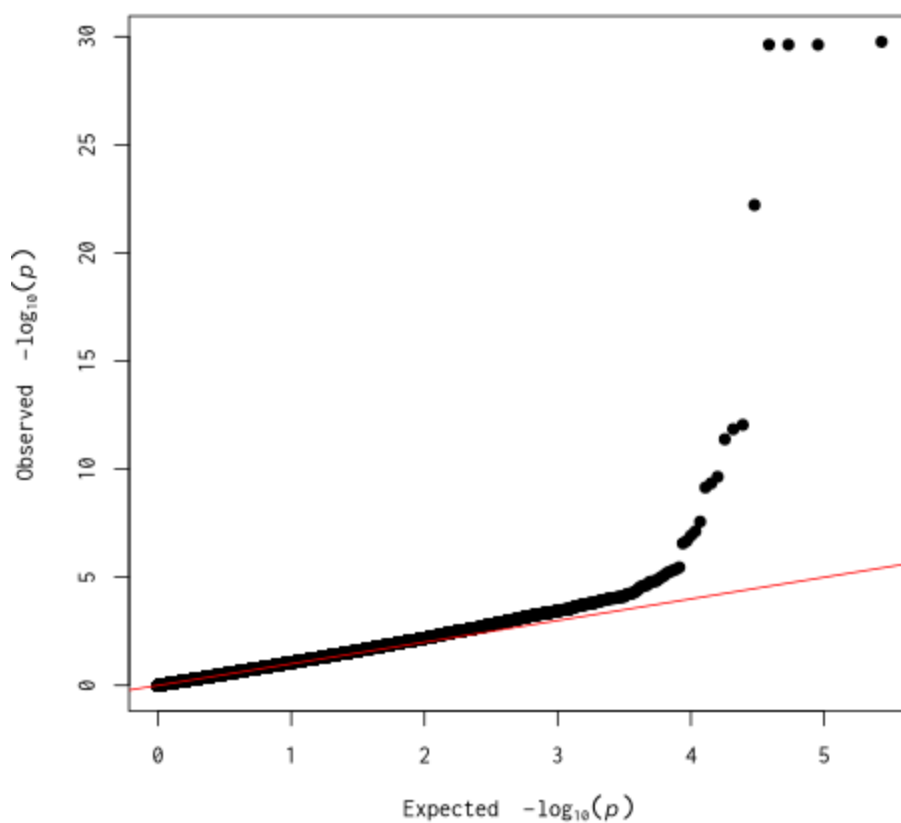
names(new4) <- c("SNP", "CHR", "BP", "P")

png("output/qq4.png")
qq(new4$P)
dev.off()
IRdisplay::display_png(file = "output/qq4.png")

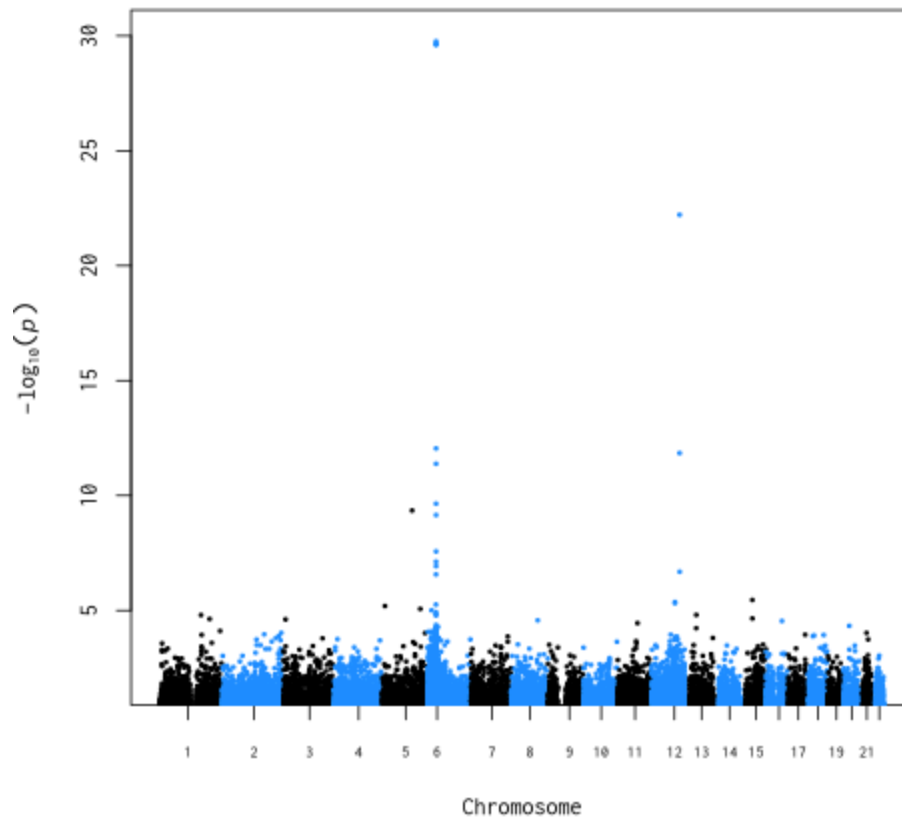
png("output/mh4.png")
manhattan(new4, pch = 20, suggestiveline = F, genomewideline = F, ymin = 2,
          cex.x.axis = 0.65, colors = c("black", "dodgerblue"), cex = 0.5)
dev.off()
IRdisplay::display_png(file = "output/mh4.png")

```

pdf: 2



pdf: 2

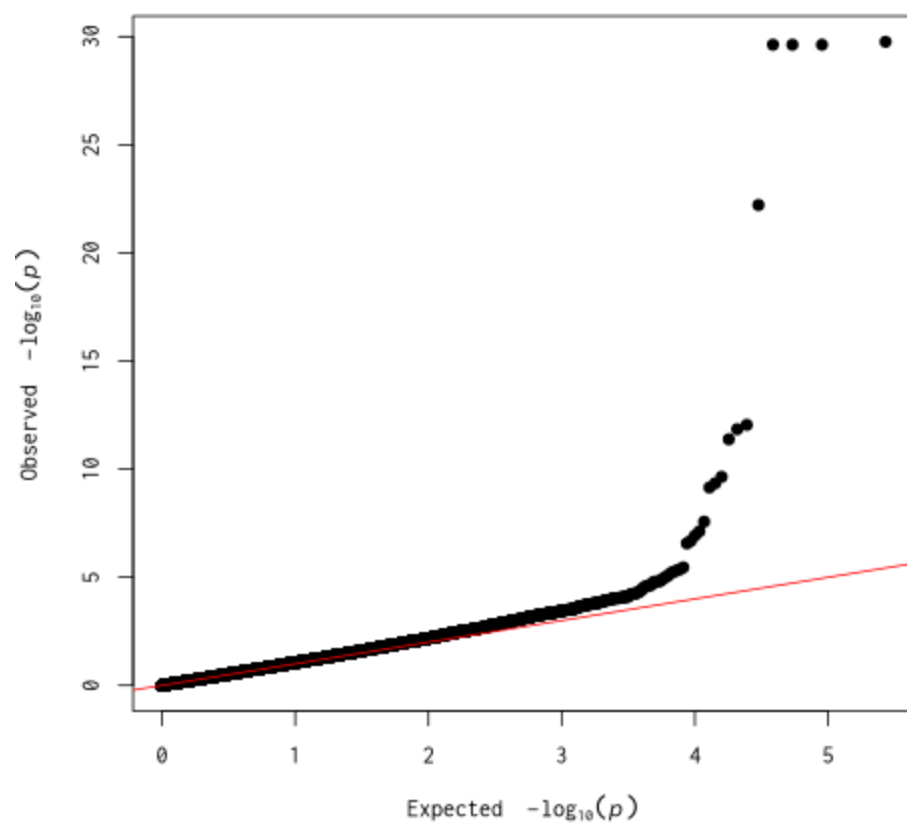


```
In [16]: # res5: newsparsegrm (partitioned/merged version) + PC covariates
new5 <- data.frame(res5$SNP, res5$CHR, res5$POS, res5$P)
names(new5) <- c("SNP", "CHR", "BP", "P")

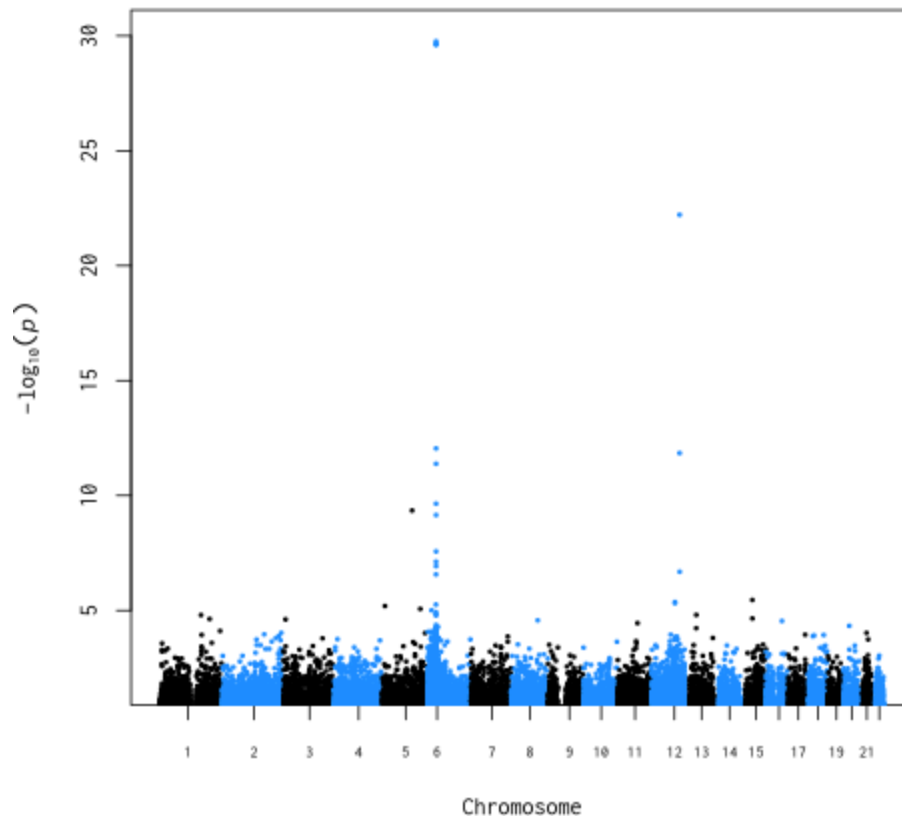
png("output/qq5.png")
qq(new5$P)
dev.off()
IRdisplay::display_png(file = "output/qq5.png")

png("output/mh5.png")
manhattan(new5, pch = 20, suggestiveline = F, genomewideline = F, ymin = 2,
  cex.x.axis = 0.65, colors = c("black", "dodgerblue"), cex = 0.5)
dev.off()
IRdisplay::display_png(file = "output/mh5.png")
```

pdf: 2



pdf: 2

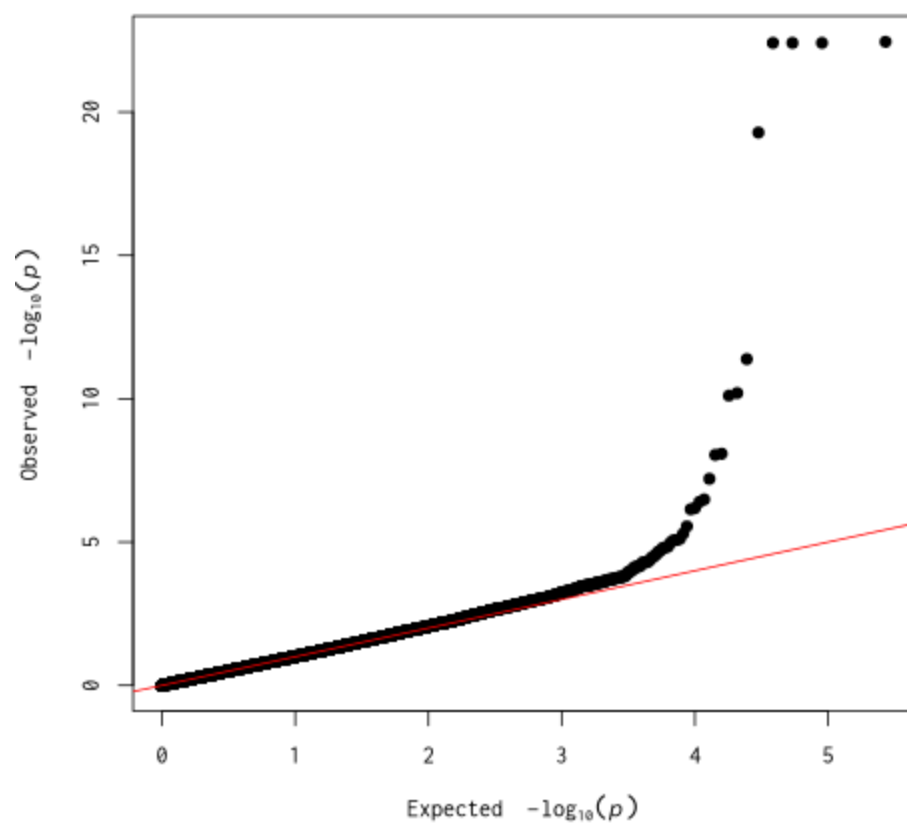


```
In [17]: # res6: sparse GRM, no covariates (forced mixed model)
new6 <- data.frame(res6$SNP, res6$CHR, res6$POS, res6$P)
names(new6) <- c("SNP", "CHR", "BP", "P")

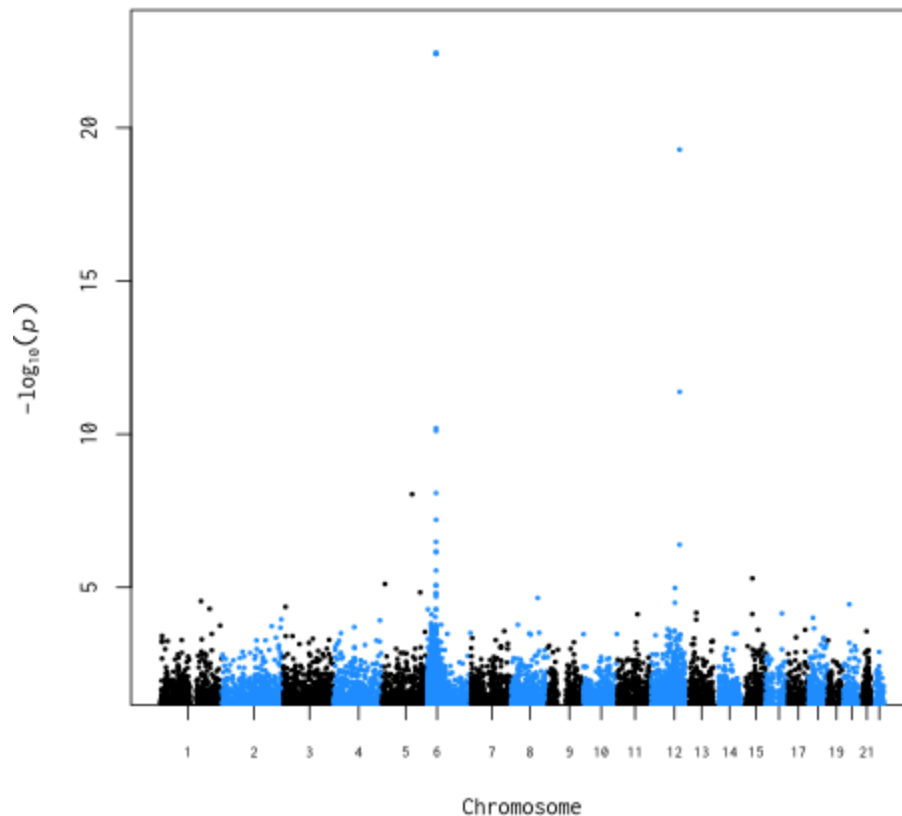
png("output/qq6.png")
qq(new6$P)
dev.off()
IRdisplay::display_png(file = "output/qq6.png")

png("output/mh6.png")
manhattan(new6, pch = 20, suggestiveline = F, genomewideline = F, ymin = 2,
  cex.x.axis = 0.65, colors = c("black", "dodgerblue"), cex = 0.5)
dev.off()
IRdisplay::display_png(file = "output/mh6.png")
```

pdf: 2



pdf: 2

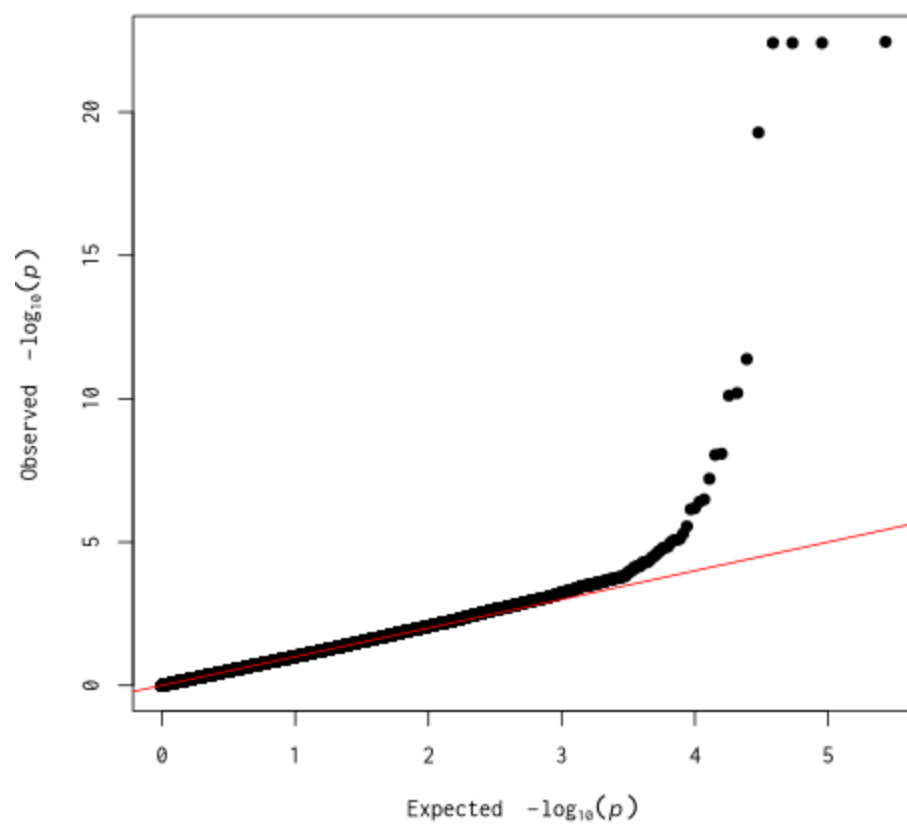


```
In [18]: # res7: newsparsegrm, no covariates (forced mixed model)
new7 <- data.frame(res7$SNP, res7$CHR, res7$POS, res7$P)
names(new7) <- c("SNP", "CHR", "BP", "P")

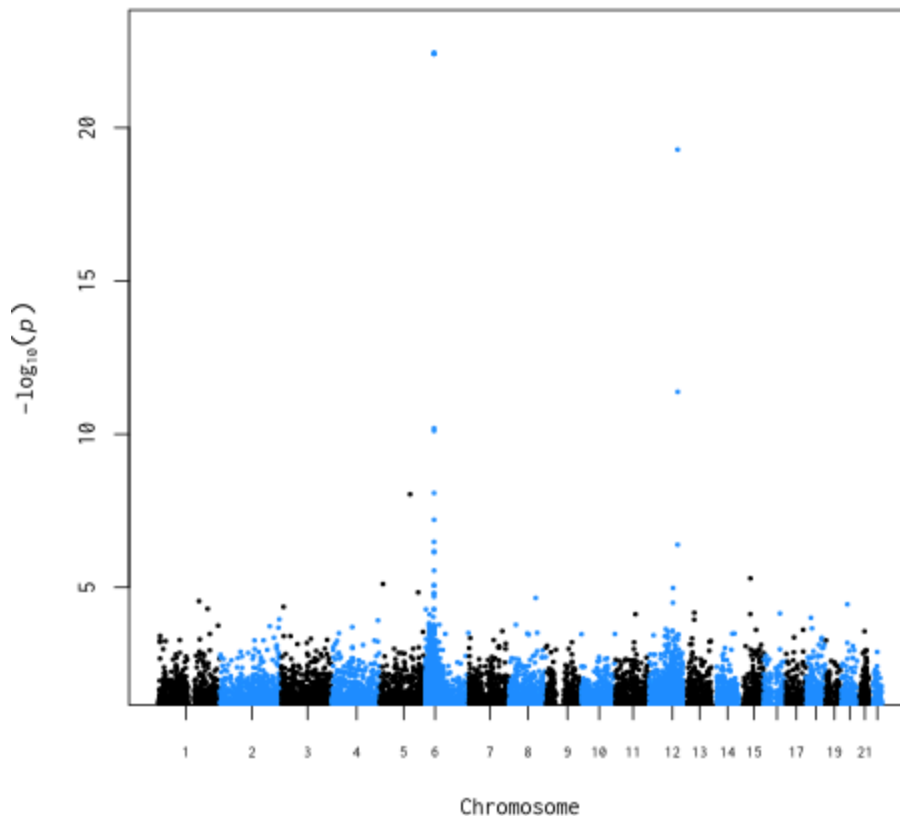
png("output/qq7.png")
qq(new7$P)
dev.off()
IRdisplay::display_png(file = "output/qq7.png")

png("output/mh7.png")
manhattan(new7, pch = 20, suggestiveline = F, genomewideline = F, ymin = 2,
  cex.x.axis = 0.65, colors = c("black", "dodgerblue"), cex = 0.5)
dev.off()
IRdisplay::display_png(file = "output/mh7.png")
```

pdf: 2



pdf: 2

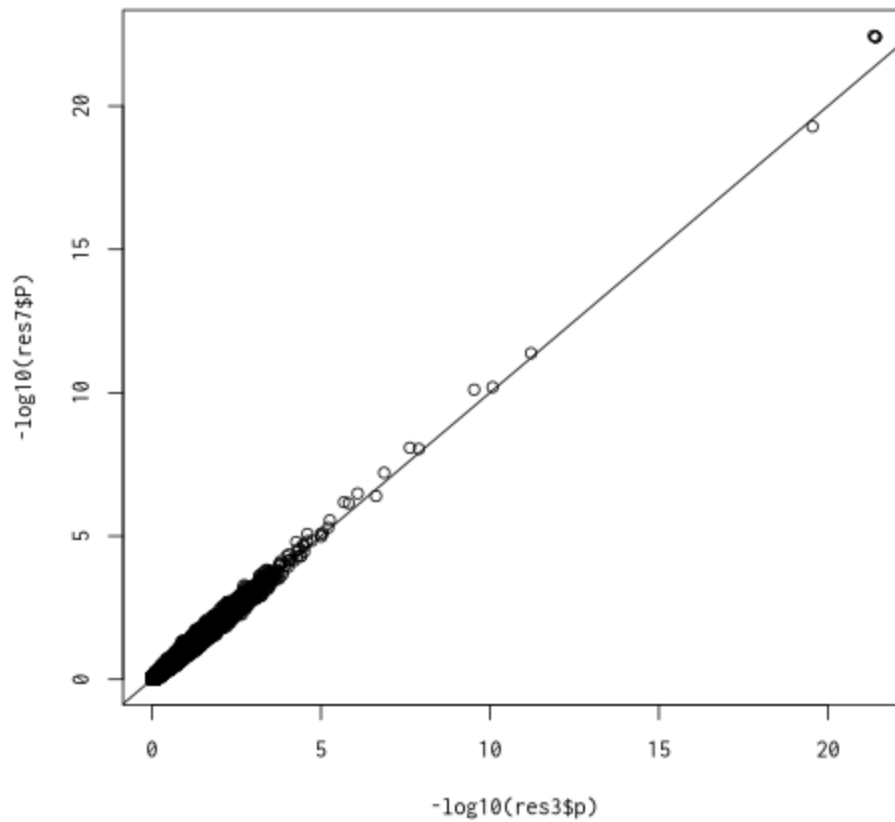


Comparing GCTA fastGWA to GCTA `--mlma`

Finally, compare the fastGWA linear-mixed-model result (sparse GRM, recalculated version, no PC covariates) against the original `--mlma` result. Expect these to line up extremely closely.

```
In [19]: # As above: points along the y = x line indicate the two methods agree closely
png("output/compareGCTAfastGWA.png")
plot(-log10(res3$p), -log10(res7$P))
abline(0, 1)
dev.off()
IRdisplay::display_png(file = "output/compareGCTAfastGWA.png")
```

pdf: 2



Other packages

- Comparable to FaST-LMM: EMMAX, GEMMA, MMM, GenABEL, Mendel
- Comparable to GCTA: DISSECT